

Quirks and All: unlocking comprehensive biological discovery out of your spectral flow cytometry datasets

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UMGCC Flow Cytometry Shared Resource

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Slides

Spectral Flow Cytometry is amazing!!!

- ❑ Capable of resolving remarkably similar fluorophores → **increased number of markers** per panel compared to conventional flow cytometry (CFC)
- ❑ Allows for rapid acquisition of millions of events in a few minutes → typically **greater number of acquired cells** per sample compared to mass cytometry (MC)
- ❑ Spectral Flow Cytometry (SFC) datasets have an unique combination of **width and depth** compared to preceding technologies



SFC has expanded our biological discovery potential

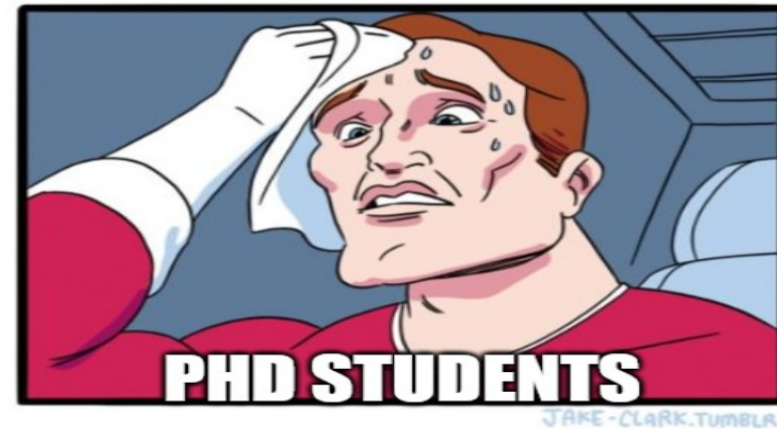


- ▶ Able to efficiently acquire greater number of cells → more thorough profiling of **rare cell subsets**
 - ▶ Build one panel, evaluate multiple cell types → invaluable for studies with **limited biospecimen**
-
- ▶ Reduced experimental cost
 - ▶ Expand our knowledge about immune system heterogeneity

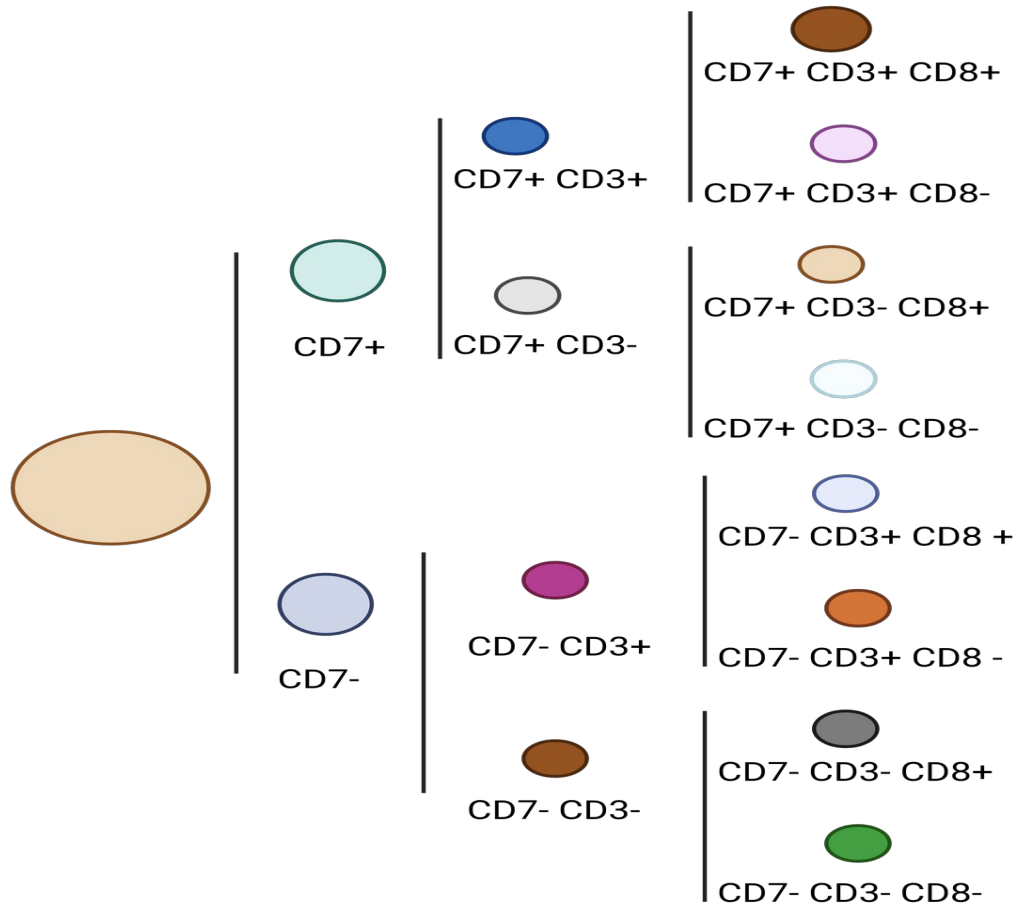
Trade-off: More complicated data analysis

- If you are only interested in a **single population** of cells → analysis process via manual gating remains essentially unchanged, just more gates

- If your goal is to profile **all cell populations** present in your .fcs →
- the breadth and depth necessitate new analytical approaches beyond manual gating



Complex Datasets: Combinatorially Exponential Clusters



- If for a given marker in your panel, you **split** the cells into **positive** and **negative** populations
- And this splitting process was **repeated for every marker** in your panel
- And at the end, we **grouped cells** that only **expressed the same markers**

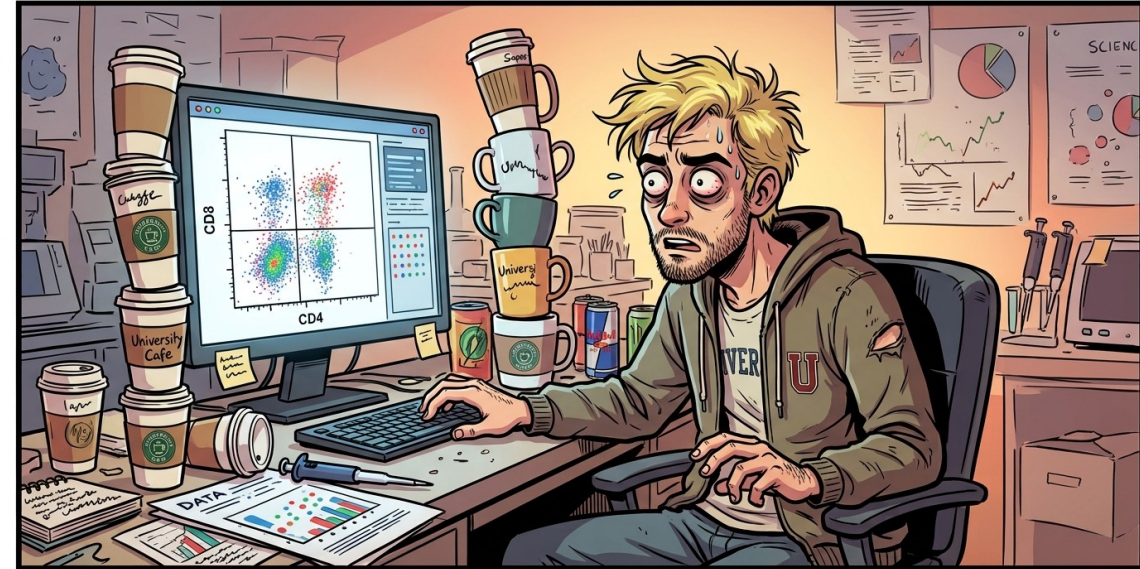
- **9-color** CFC panel → 512 groups at most
- **40-color** SFC panel → 1.1e+12 groups (**trillion**)
- **60-color** SFC panel → 1.2e+18 groups (**quintillion**)

If I manually checked every combination

- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

- 1 gate, every 10 seconds → 6 groups/minute
- 60 minutes in an hour → 360 groups/hour
- Awake 16 hours a day → 5760 groups/day
- 365 days per year → **2,102,400 groups/year**

- 76 years - 22 pre-PhD = 54 years
- → **1.13e+8 groups/life-time**



Continuously since the Big Bang



- **60-color SFC panel** → **$1.2\text{e}+18$ groups (quintillion)**

- 365 days per year → **2,102,400 groups/year**
- Invent a time-machine, become immortal
- 13.8 billion years → **$2.9\text{e}+16$ groups/reality**

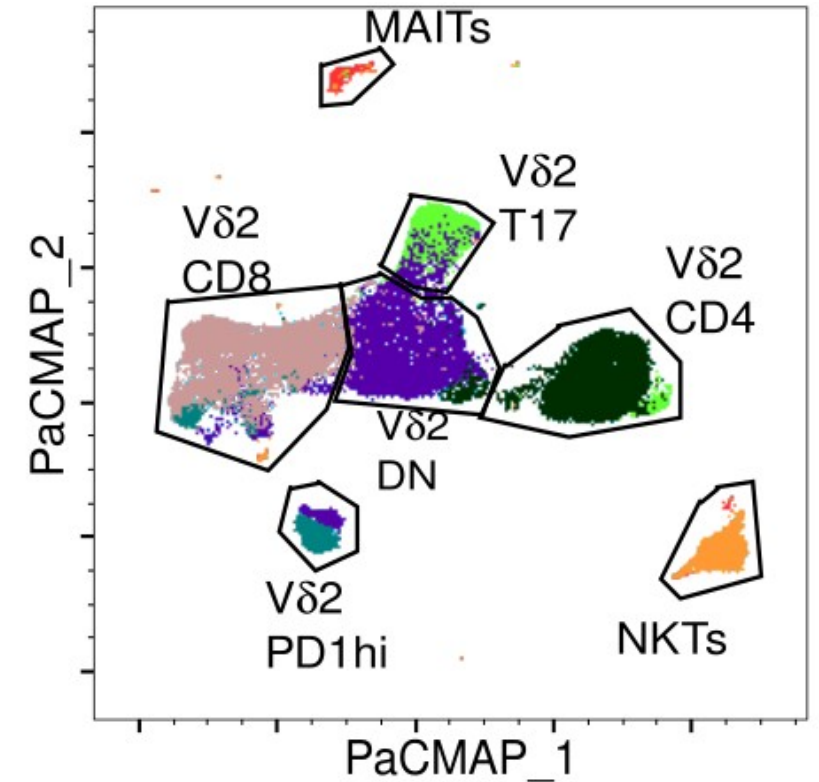
- So we need ~ 50 immortal researchers working continuously since the Big Bang just to check all group combinations for a **single 60-fluorophore panel .fcs file**

Alternative: Unsupervised Analysis (Algorithms)

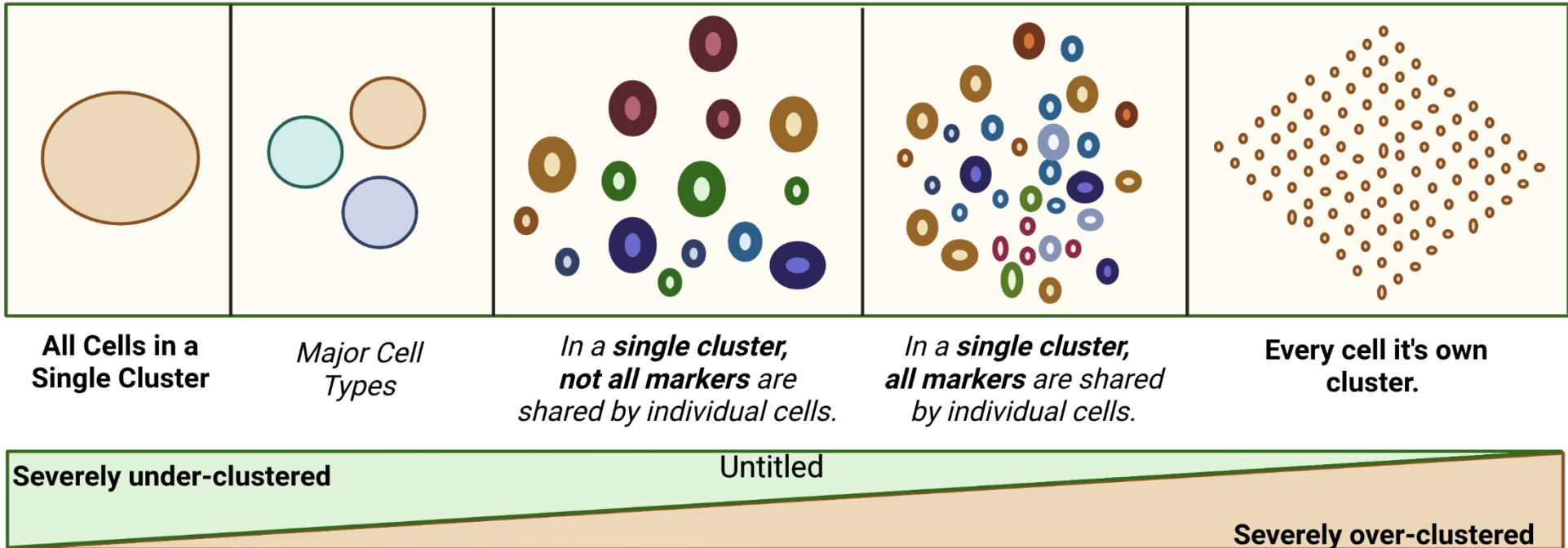
- Consequently, if the goal is to characterize all cell populations present, some form of **unsupervised (algorithm) or semi-supervised** analytical approaches will be needed

- **Clustering** → FlowSOM, Phenograph, Xshift, etc.
- **Dimensionality visualization** → opt-SNE, UMAP, PaCMAP, Phate, etc.

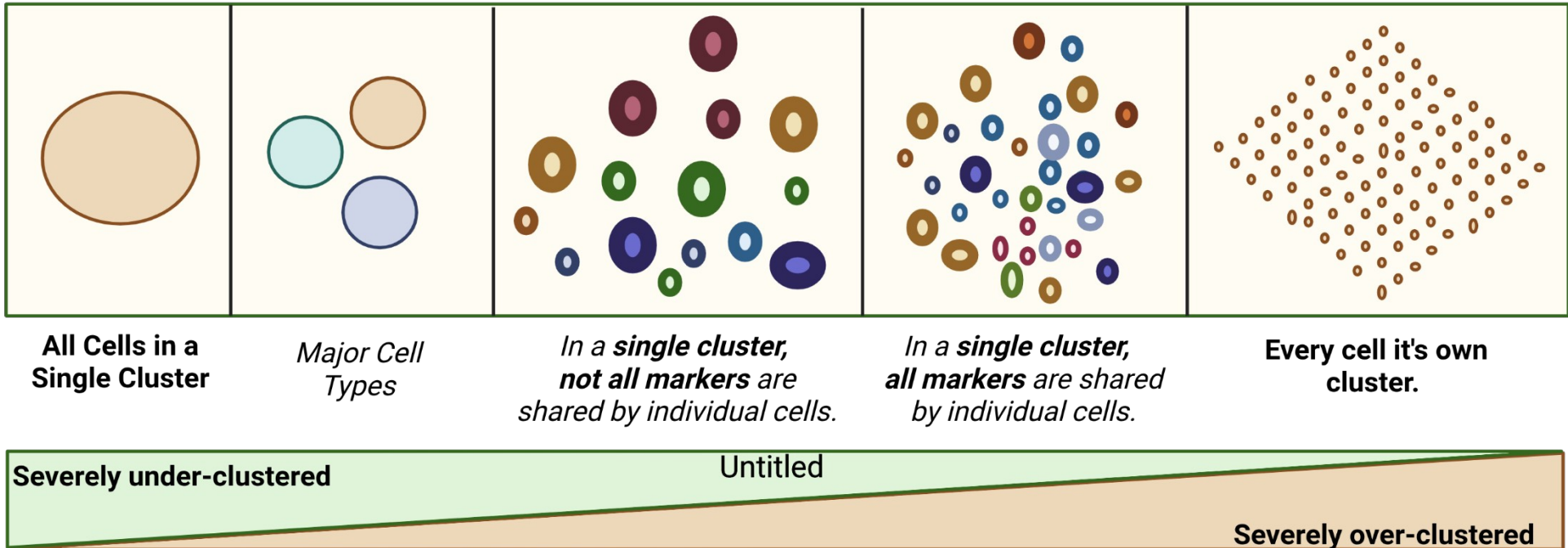
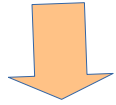
- **Automated Gating** → openCyto, flowMagic, ElasticGate, etc.



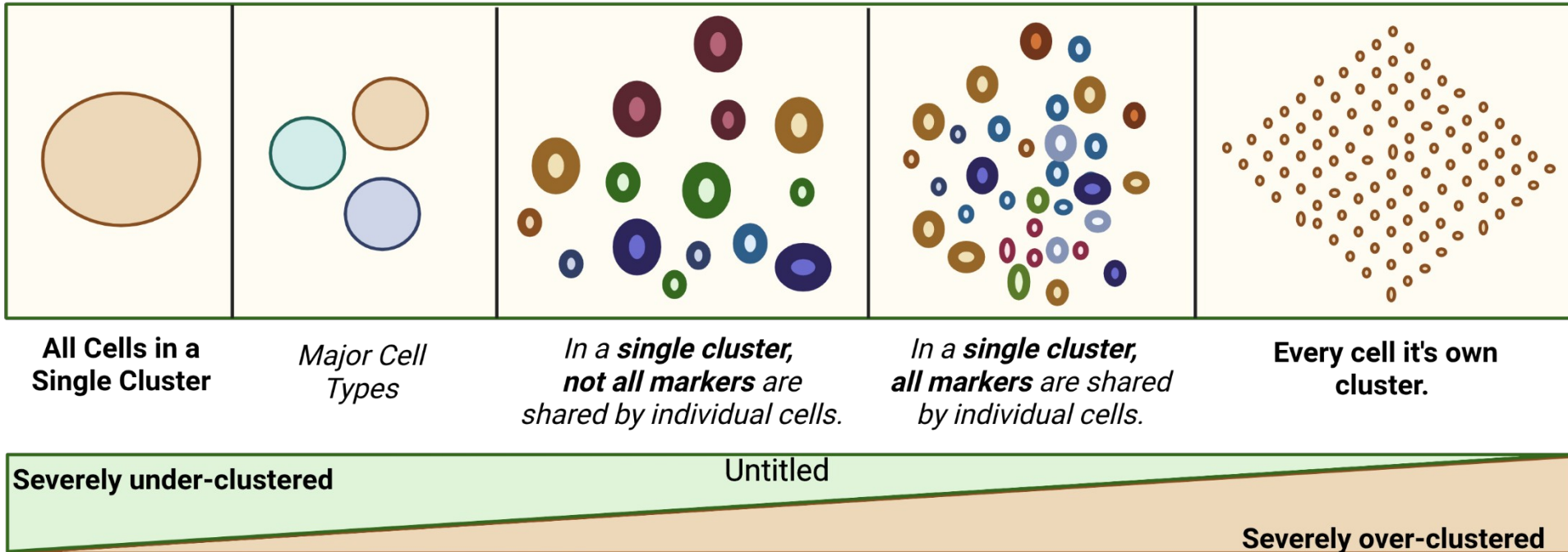
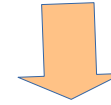
What all the arguments are about



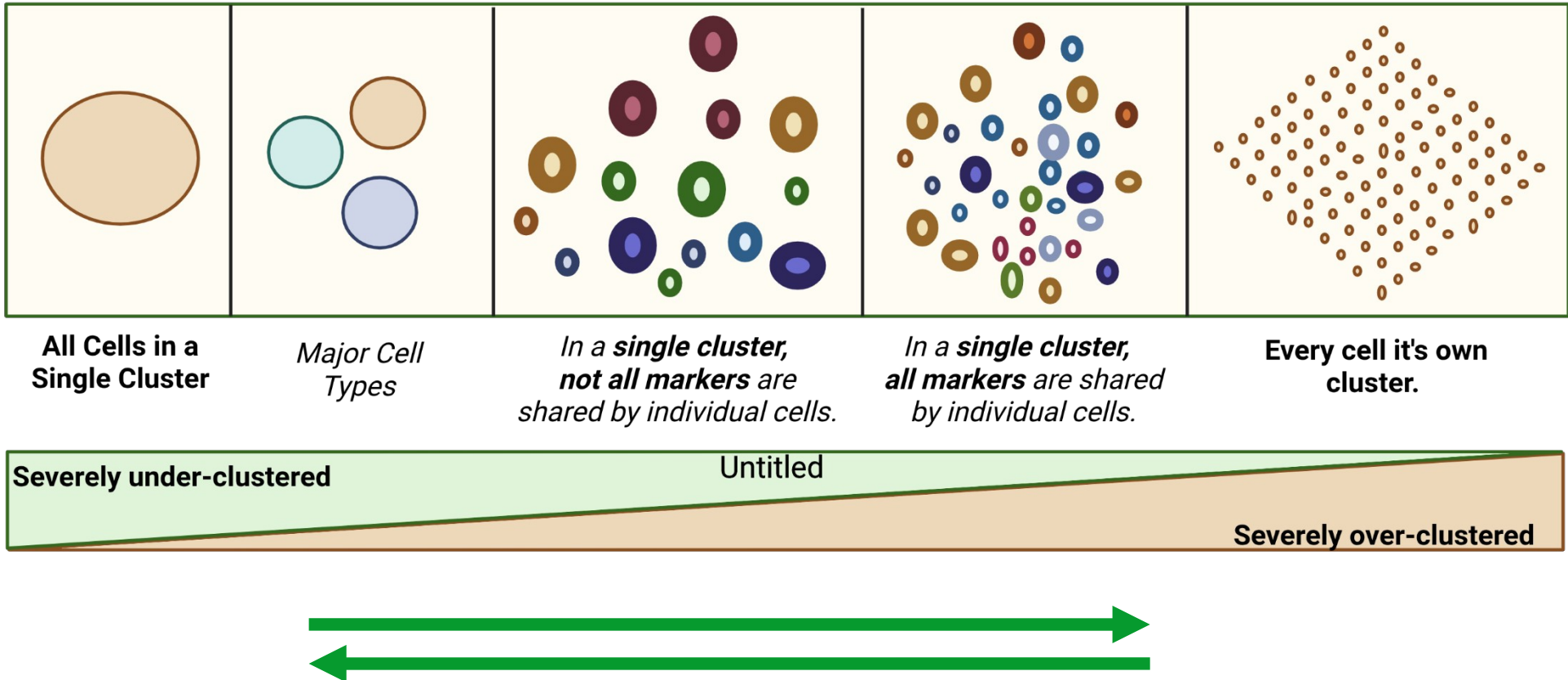
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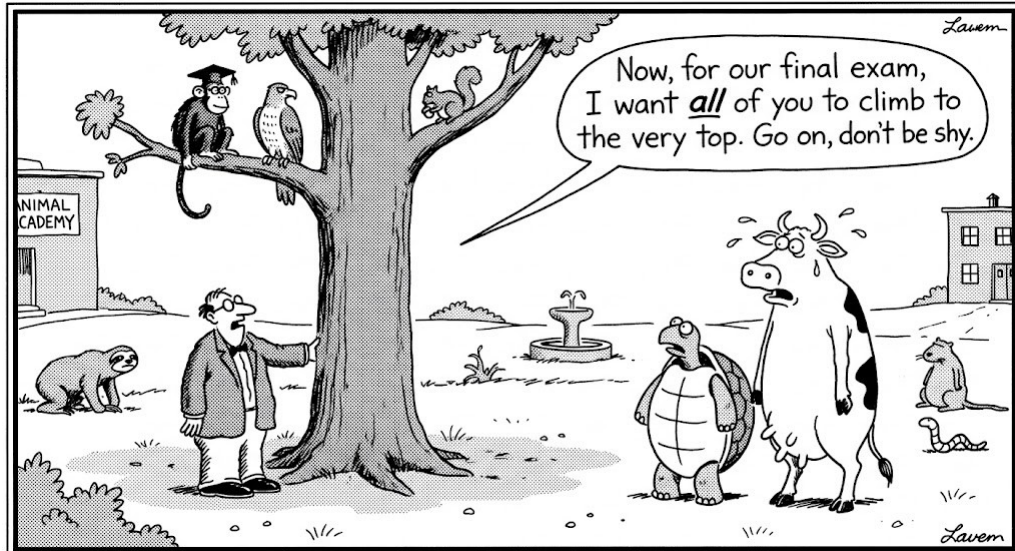
What all the arguments are about



What all the arguments are about



Adapted algorithmic tools struggle with number of events



- Many of these software tools were originally implemented as R or Python packages, for use in analysis of mass cytometry or single-cell RNA-seq datasets

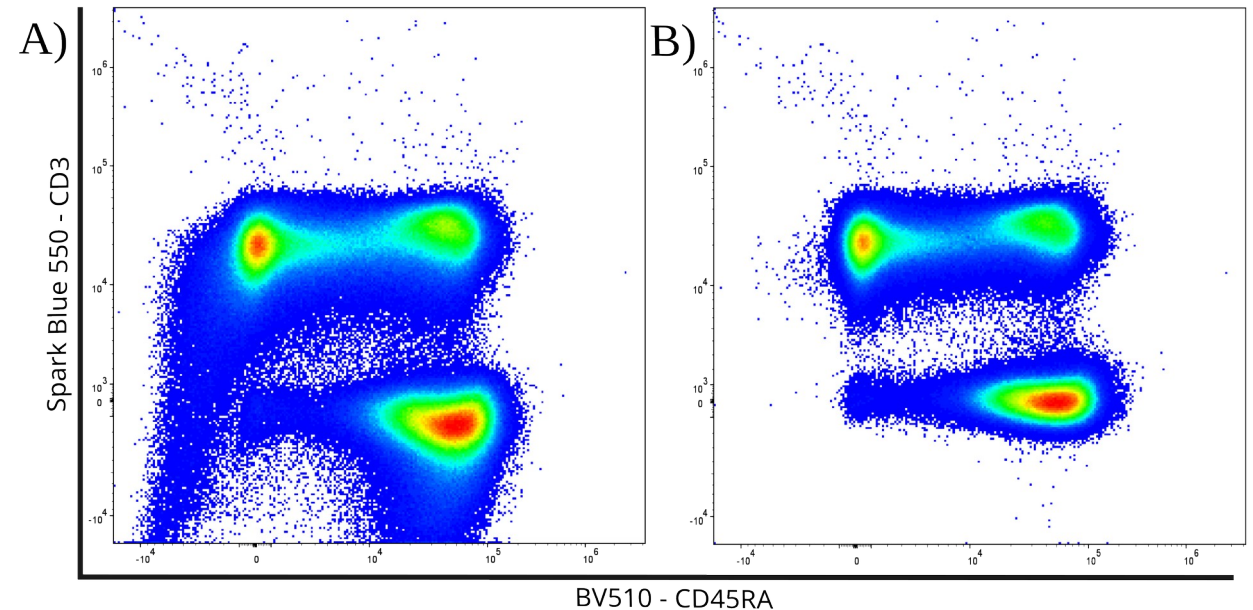
- Mass cytometry → similar number of markers, typically fewer events, no unmixing variation due to autofluorescence

- Single-cell RNA-seq → substantially more markers, typically way fewer events. No equivalent “negative” gate events for not expressed transcripts

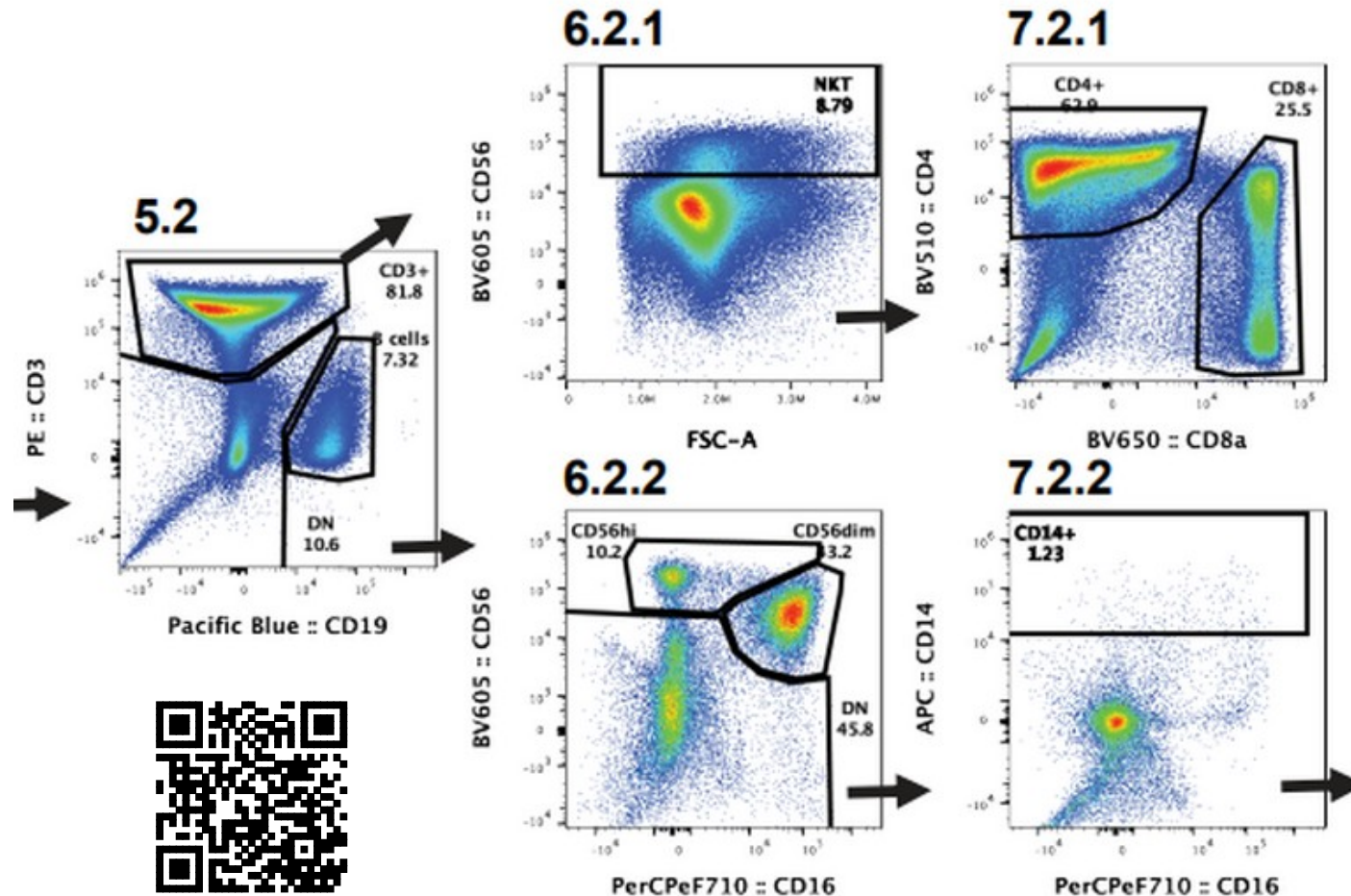
- Down-sampling becomes often necessary

Adapted algorithmic tools struggle with unmixing errors

- Adapted software packages heavily rely on having a consistent **median fluorescent intensity (MFI)** to cluster cells based on positive or negative marker expression
- Existing **batch correction** / normalization / scaling tools target instrumental or **experimental** level variation for correction
- Unmixing errors** can seriously shift the MFI ranges for “negative” and “positive” cells for **individual** acquired specimens



“We discovered a novel dendritic cell cluster”



- Despite the critical importance unmixing controls play, few tools permit us to determine quality of the individual control before unmixing

- Existing quality control algorithms (PeacoQC, FlowAI, etc.) primarily focus on detecting laser and fluidic issues in already acquired .fcs files.

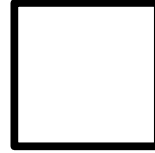
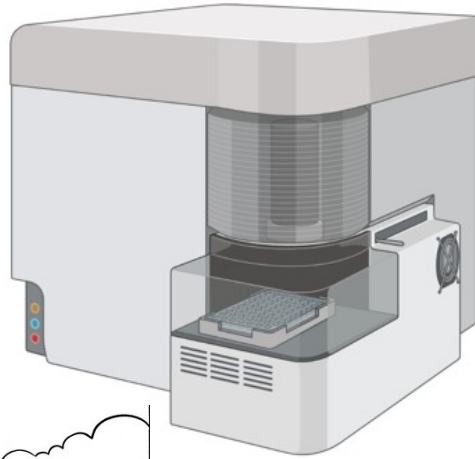
- Unmixing errors can be introduced by tandem degradational, additional autofluorescences, undiagnosed instrumental issues, etc.

Full biological discovery potential often out of reach

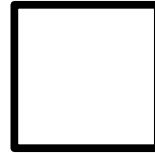


- Most flow cytometrist don't code
- You likely lost your bioinformatician while describing manual gating
- The statisticians were never consulted
- For **most** spectral flow cytometry datasets, only a **handful of cell populations** end up being analyzed by any method

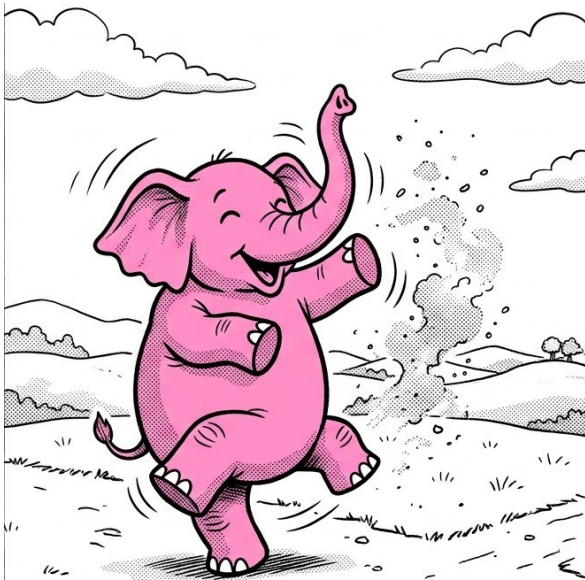
To unlock the comprehensive biological discovery potential, we need to address these pink elephants



- Quality of the unmixed full-stained data is linked to the quality of our single-color controls (and our ability to provide appropriate autofluorescence controls).



- If we are acquiring 3 million cells to detect 100 rare cells, how do we thoroughly explore the rest of the sample to avoid wasting useful data?



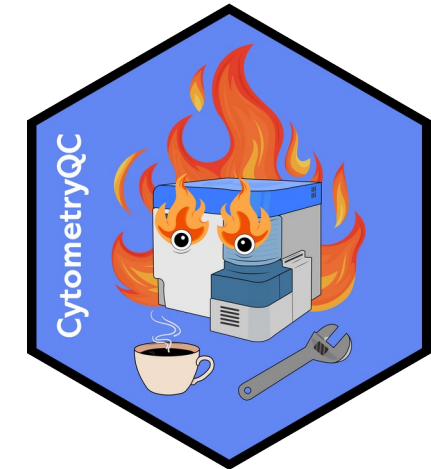
Building open-source tool-sets to address SFC's quirks



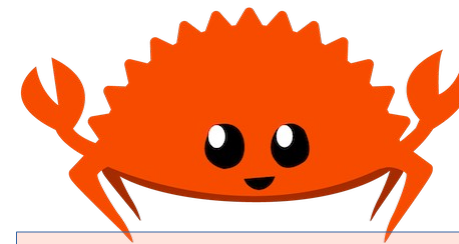
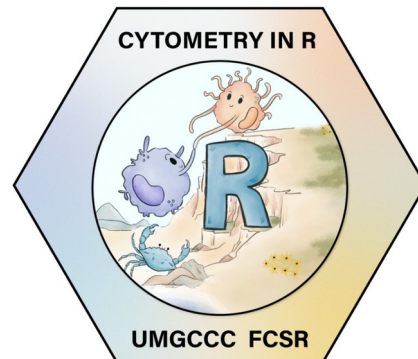
Spectral Signature
Characterization and Isolation



Incorporation of manually-
annotated gating information

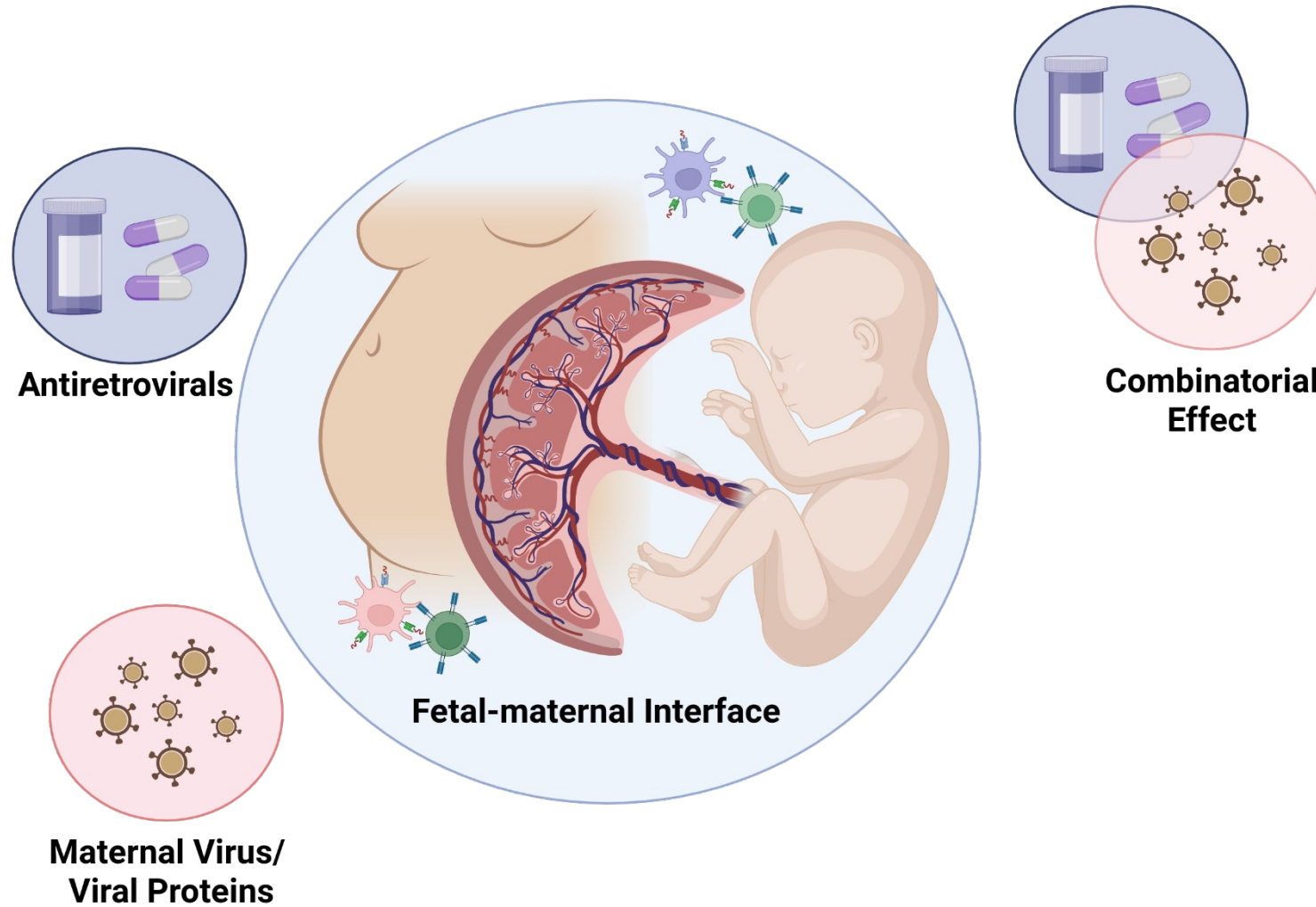


Automated Instrument
Quality-Control Dashboards



Rust

My PhD's focus: HIV-exposed Uninfected Infants



HIV-exposed Uninfected Infants

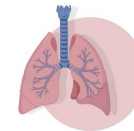


HIV-unexposed Uninfected (HUU) Infants

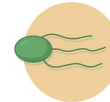
HIV-exposed Uninfected (HEU) Infants



↑ Respiratory Infections



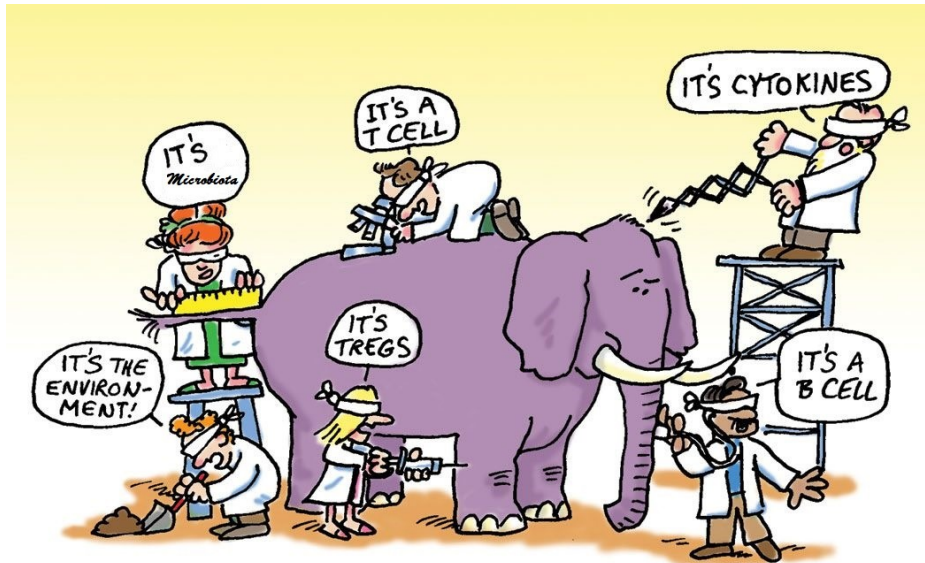
↑ Enteric Infections



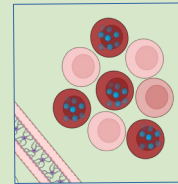
↑ Hospitalization



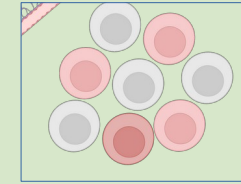
Is there an immunological cause?



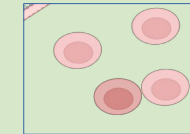
□ Immunological cause?



Too
active



Too
inactive

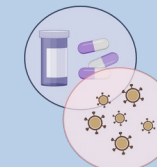


Cells
missing

Other factor?

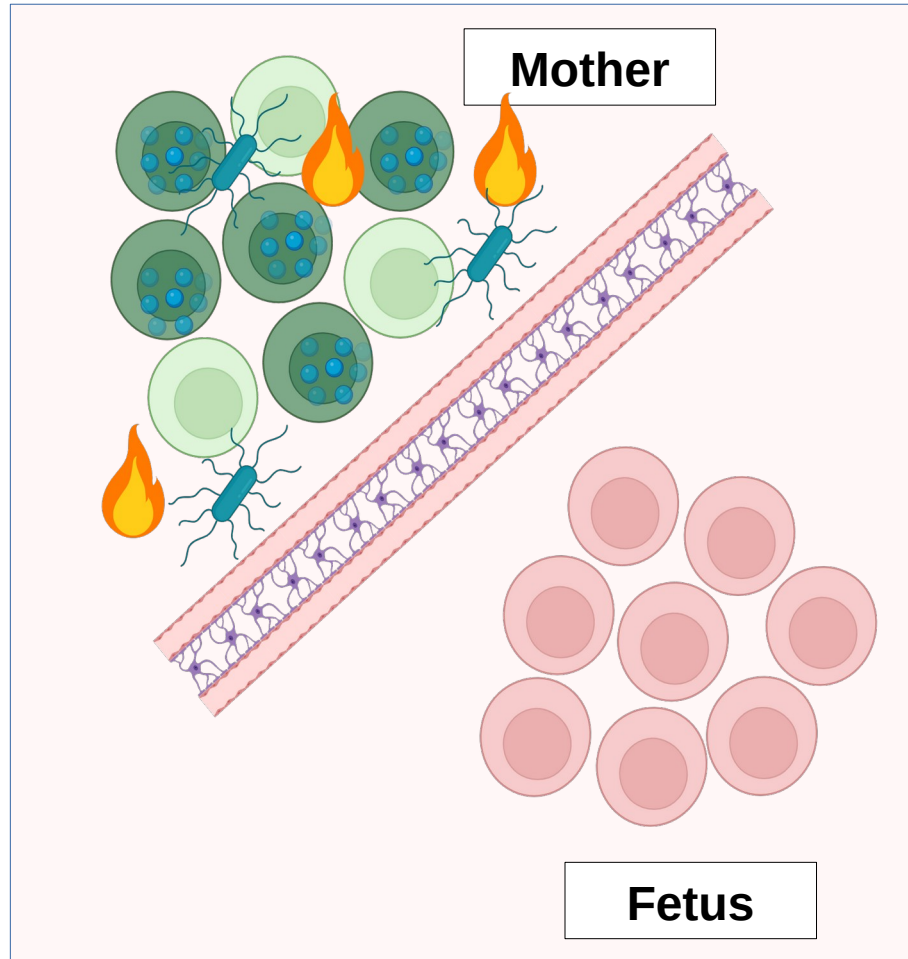


Formula vs.
Breastfeeding



Maternal
Antiretrovirals

Infection during pregnancy



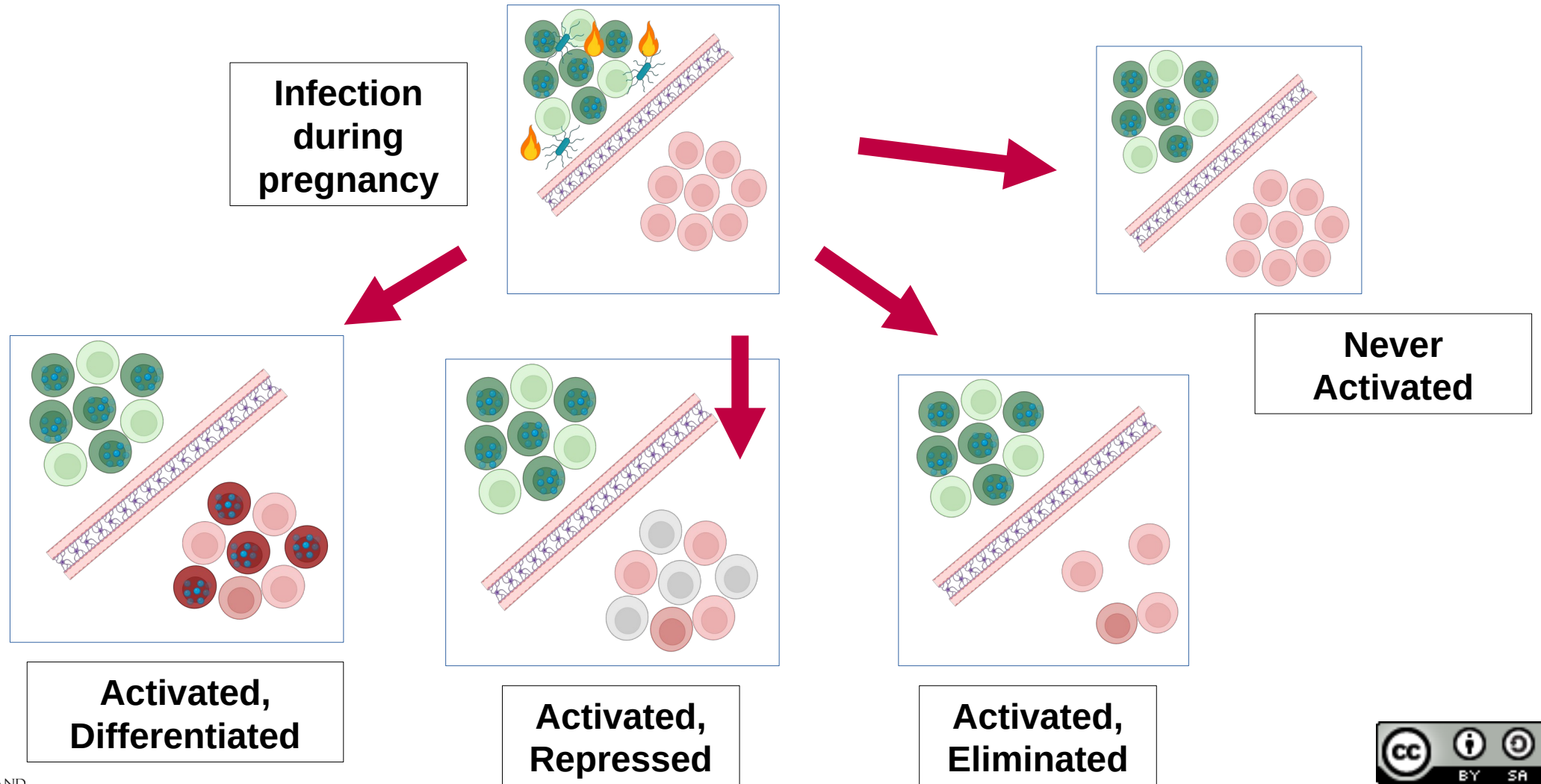
Does the maternal immune system respond?

- Is there interaction?
- Is there damage?

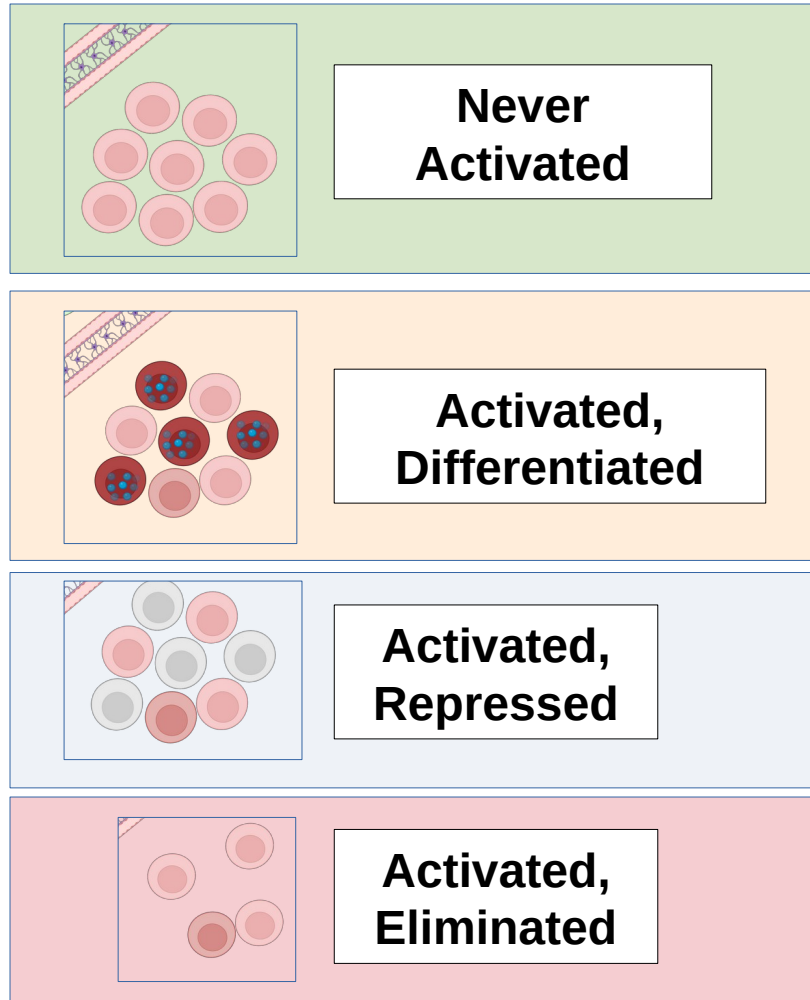
Is the developing fetal immune system affected?

- Shared circulation
- Location, Scale, Timing
 - Cold, Upper Nasal
 - Influenza, Systemic

What effect does early exposure have on the fetal immune system?



Does this fetal exposure impact immune responses during early life?



Conflicting Studies

Some Studies:

- Increased frequency of Th1-like cytokines in T cells *ex-vivo*
- Decreased proliferation in culture
- Lower antibody titers for maternal antibodies



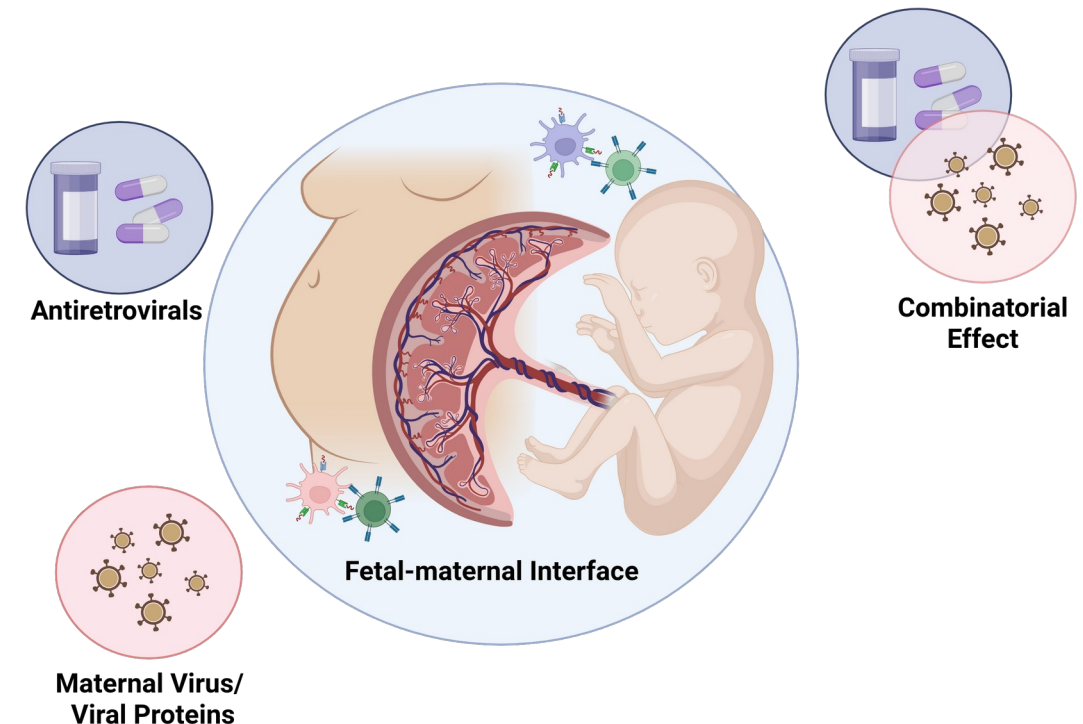
Other Studies:

- Same (or lower) frequency of Th1-like cytokines in T cells
- *ex-vivo*
- No change in proliferation in culture
- Higher antibody titers for maternal antibodies

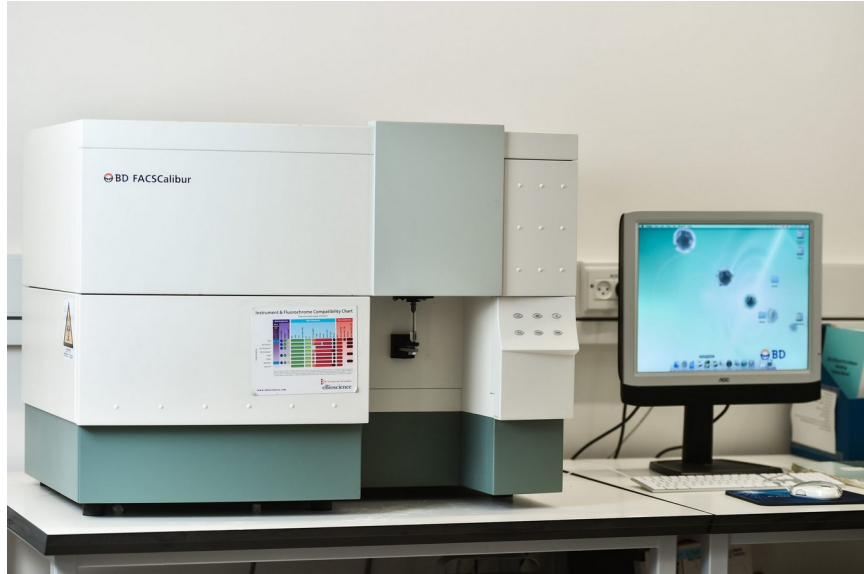
Treatment standards changed over time

- As anti-retrovirals became more available, health policy changed
- Maternal antiretroviral until birth →
- Single antiretroviral for life →
- Combined antiretroviral for life

Prior studies may have grouped infants with varied viral and antiretroviral exposures together



Snapshots in Time



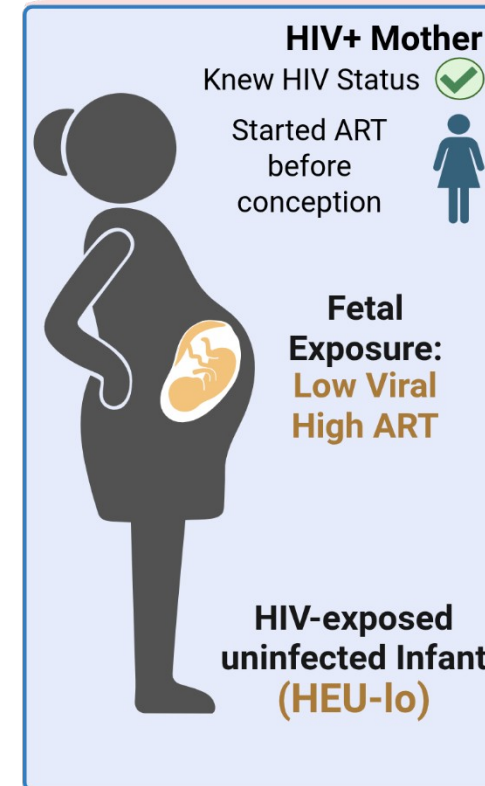
- Even carefully controlled ground-breaking studies are defined by the technology and resources available at time

- Conventional Flow Cytometers
- Calibur (2007) (4-fluorophores)
- LSR II (2017) (14-fluorophores)
- Spectral Flow Cytometers
- Up to 60! fluorophores now

Our Malawian Cohort

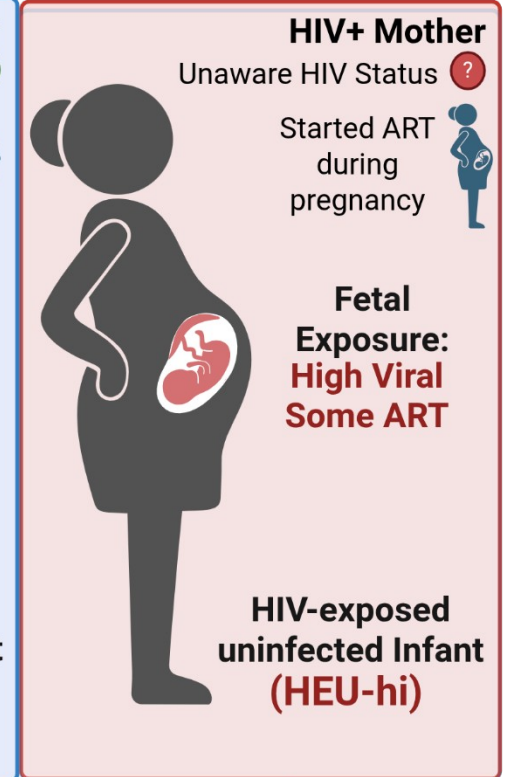


Maternal Viral Load:
Undetectable



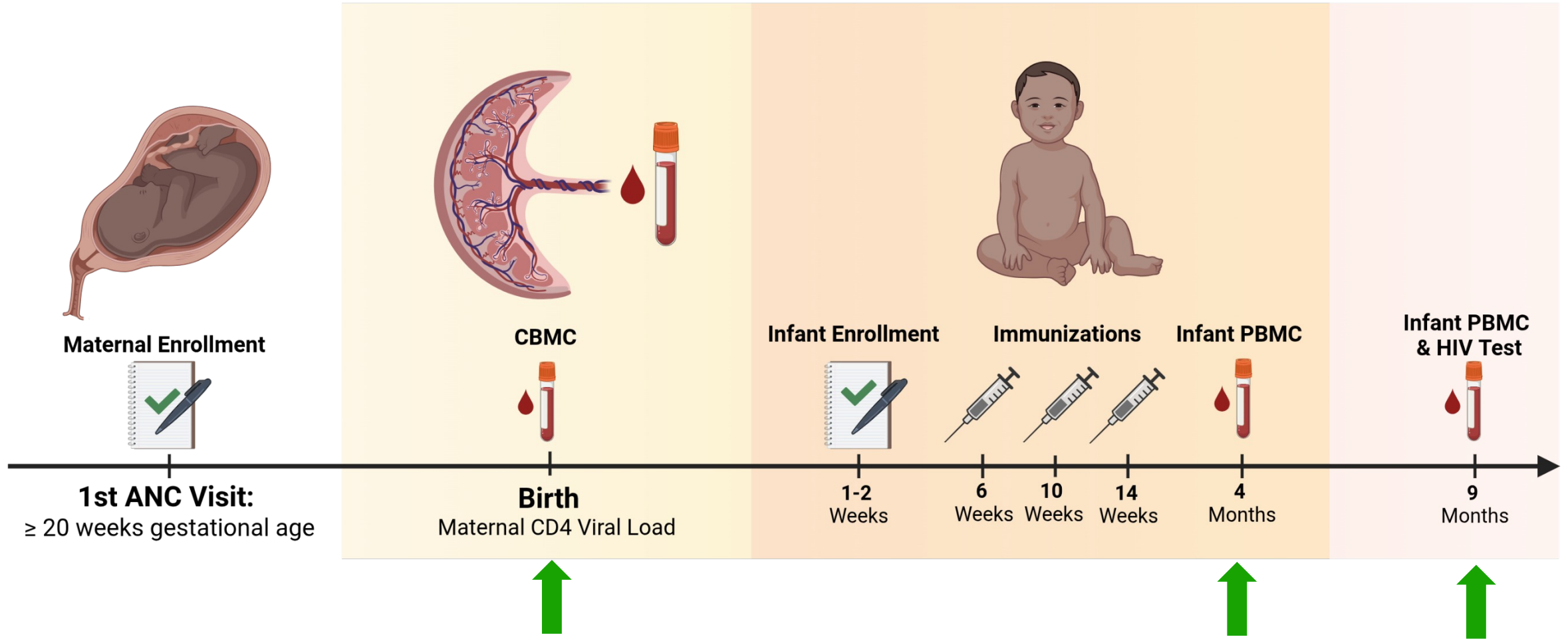
Fetal Antiretroviral Exposure: Extended

Maternal Viral Load:
Very High

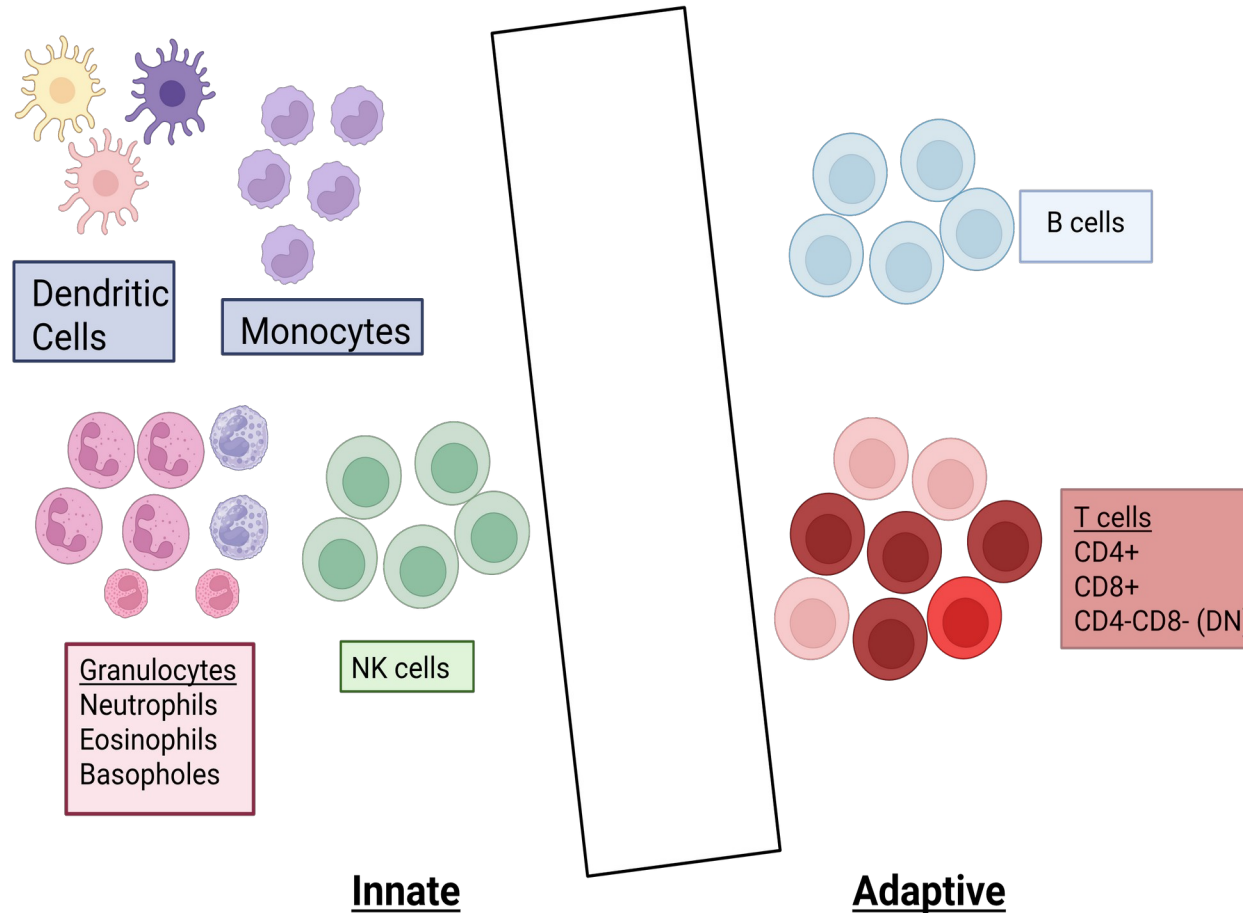


Fetal Antiretroviral Exposure: Limited

Study Timeline



Cells of the Immune System



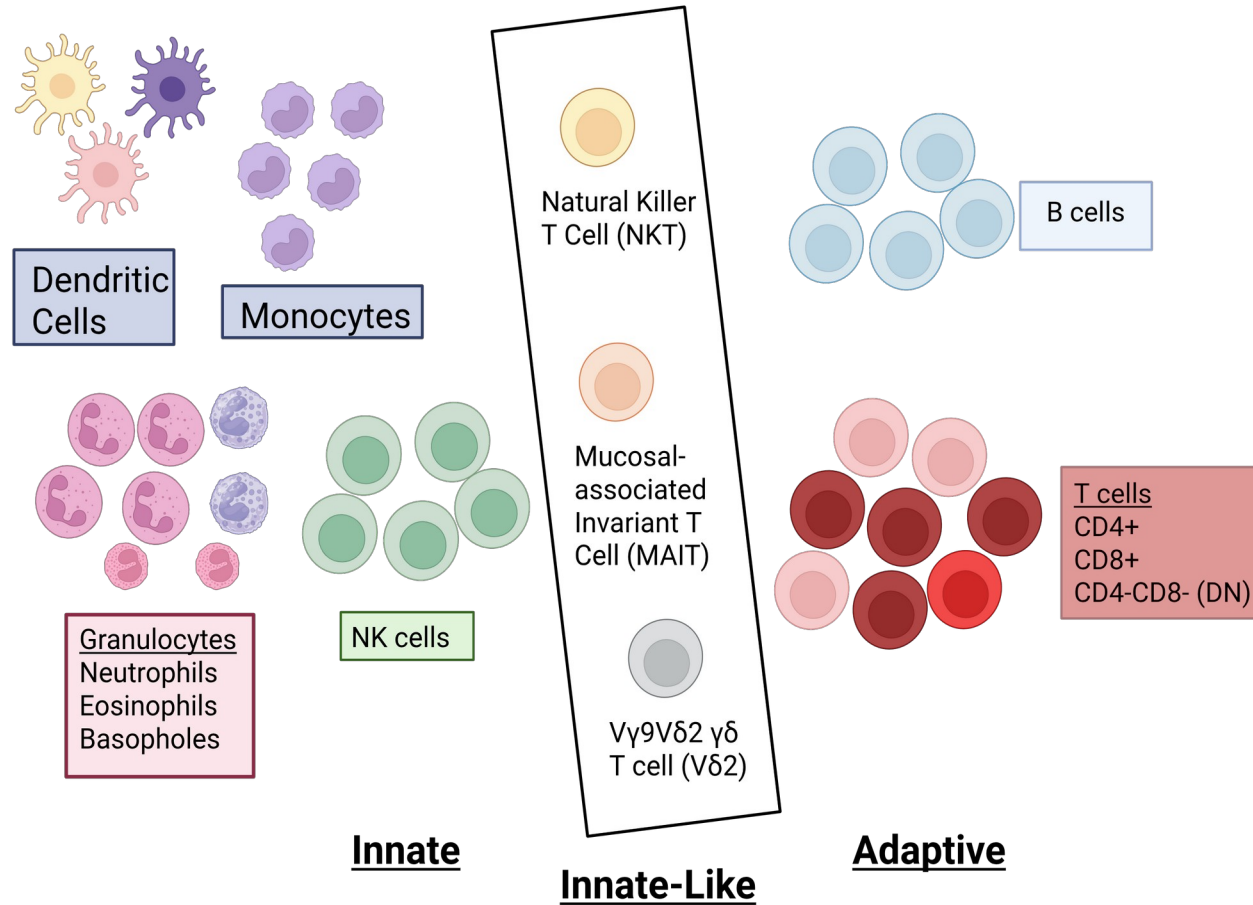
Innate:

- Rapid response
- Conserved patterns

Adaptive:

- Slower response
- Antigen-specific

Innate-like T cells



Natural Killer T cells (NKTs)

- Va24Ja18+ (antibody)
- CD1d (tetramer) with PBS-57 (Lipid)

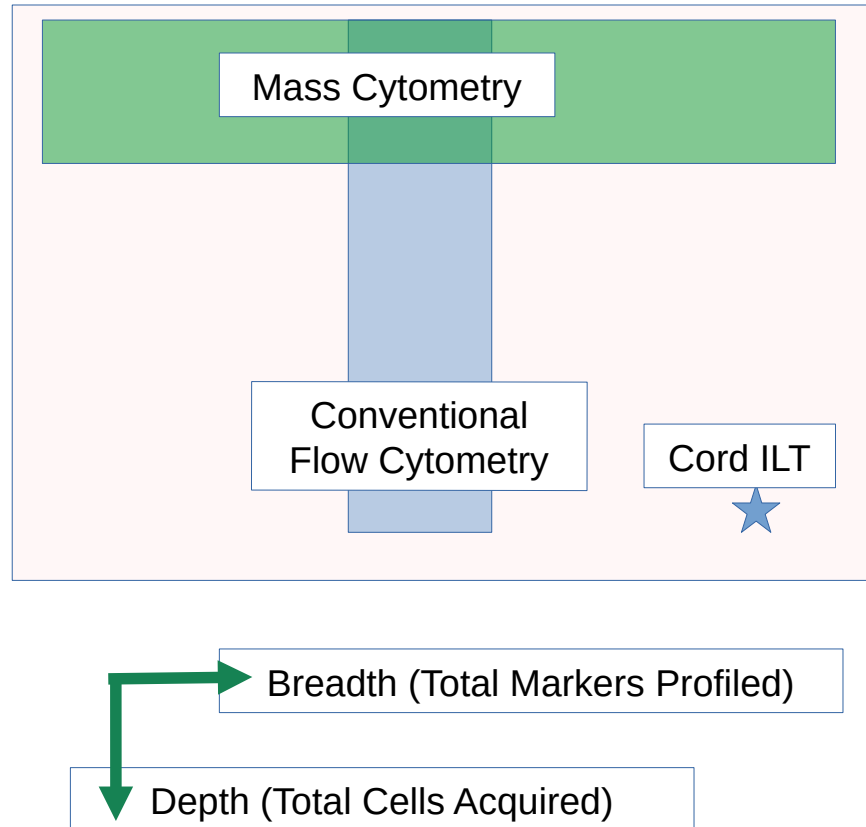
Mucosal-Associated Invariant T cells (MAITs)

- Va7.2+CD161hi (antibody)
- MR1 (tetramer) + OP-RU-56

V γ 9+V δ 2+ $\gamma\delta$ T cells (V δ 2)

- V γ 9+V δ 2+ (antibody)

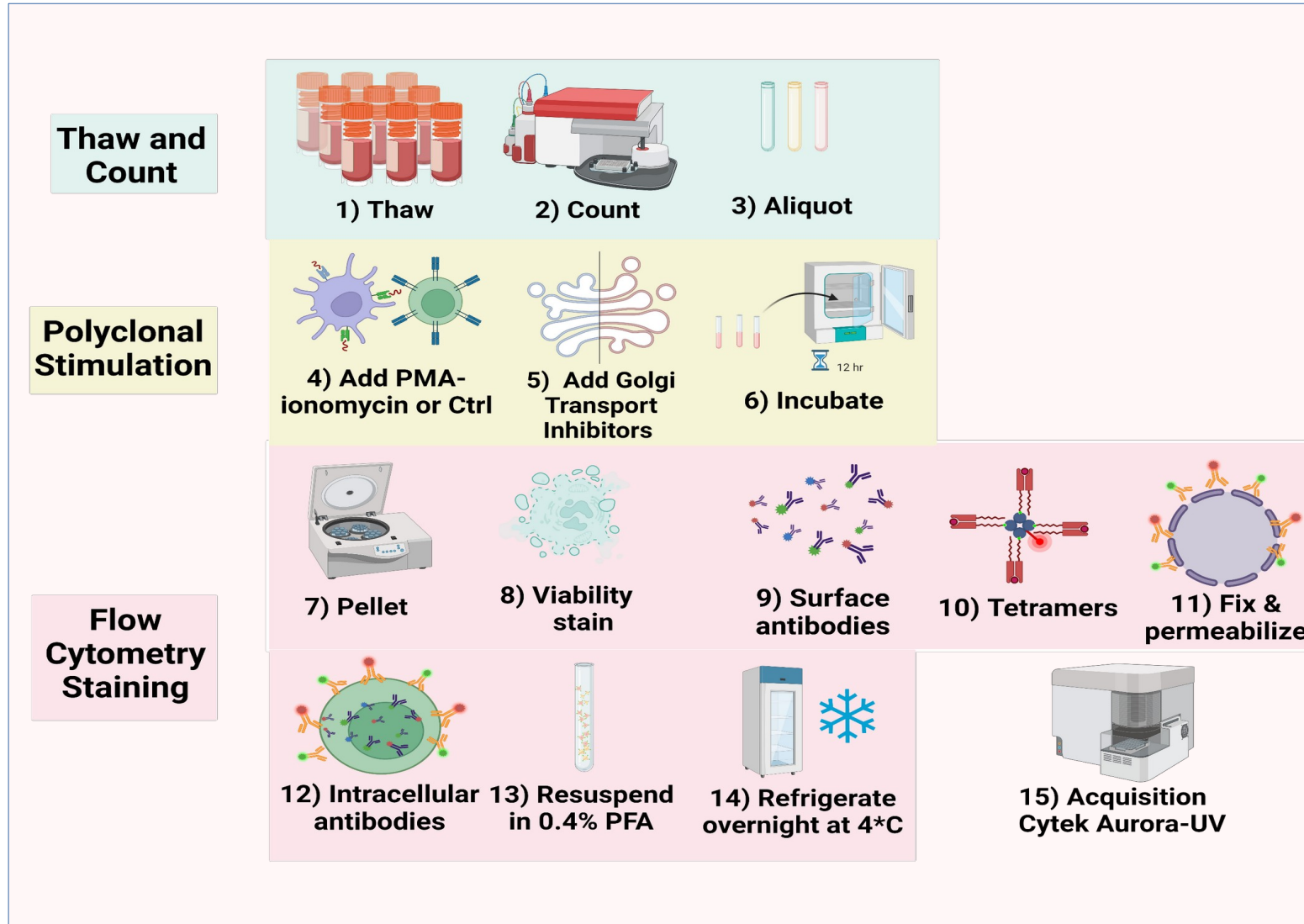
Little is known about cord blood Innate-like T cells



- Additional lineage markers
 - V δ 2+, V α 24J α 18+, V α 7.2+
- Unfeasible for Conventional Flow Cytometry due to smaller panels

- Rare in Cord Blood!
 - < 0.1% of T cells
- Unfeasible for Mass Cytometry due to low acquisition rate

Innate-like T cell SFC Panel – 29 fluorophores



32-Fluorophore Spectral Flow Cytometry Panel

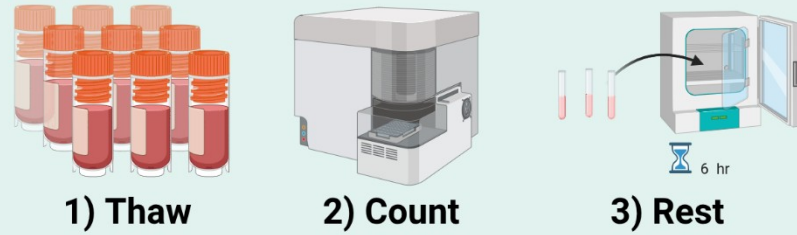
UV1	Spark UV 387		
UV2	BUV395	CD62L	(SK11)
UV6	cFluor UV440		
UV7	BUV496		
UV9	BUV563	CD69	(FN50)
UV10	BUV615	CCR4	(1G1)
UV11	BUV661	V δ 2	(B6)
UV14	BUV737	CD38	(HB7)
UV16	BUV805	CD4	(SK3)
V1	BV421/		
V2	Spark Violet 423		
V3	Pacific Blue	CD14	(M5E2)
V3	Pacific Blue	CD19	(SJ25C)
V4	SYTOX Blue		
V5	BV480	CD161	(HP-3G10)
V6	SBV515		
V7	BV510		
V8	BV570	CD16	(3G8)
V9	Qdot 585		
V10	BV605	CD45RA	(HI100)
V11	BV650	CD8	(RPA-T8)
V13	BV711	Va7.2	(3C10)
V14	BV750	IFN γ	(B27)
V15	BV786	CCR6	(11A9)
V16	LD NIR 786		

B2	Alexa Fluor 488	FoxP3	(236A/E7)
B3	Spark Blue 550	CD3	(SK7)
B4	Spark Blue 574	CD45	(2D1)
B6	RB613	PD1	(EH12.1)
B7	NovaFluorBlue660		
B8	SBB675		
B9	NovaFluorBlue690		
B10	RB705	CD26	(M-A261)
B12	RB744/		
B13	NovaFluorBlue760		
B14	RB780	CXCR5	(RF8B2)
YG1	PE	ICOS	(ISA-3)
YG2	Spark YG 593		
YG3	PE-Dazzle 594	TNF α	(MAB11)
YG4	PE-Fire 640/		
YG5	PE-Cy5	CXCR3	(G025H7)
YG6	NovaFluorYellow690		
YG7	PE-Fire 700	CD127	(A019D5)
YG8	PE-Fire 744	CD25	(M-A251)
YG9	PE-Vio 770	HLA-DR	(REA805)
YG10	PE-Fire810		
R1	APC	CD39	(A1)
R2	Alexa Fluor 647	IL2	(MQ1-
R3	cFluor R685		
R4	APC-R700	CD107a	(H4A3)
R5	NovaFluorRed725		
R6	Zombie NIR	L/D	
R7	APC-Fire 750	CD27	(O323)
R8	APC-Fire 810	CCR7	(G043H7)

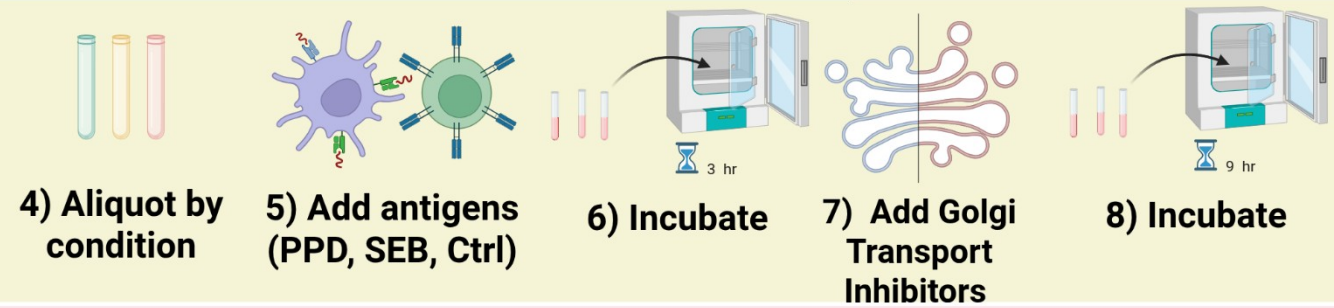
- Lineage markers for ILTs
- Phenotypic markers for T and NK cells
- Cytokines (TNF α , IFN γ , IL-2)

$\alpha\beta$ T cell SFC Panel – 32 fluorophores

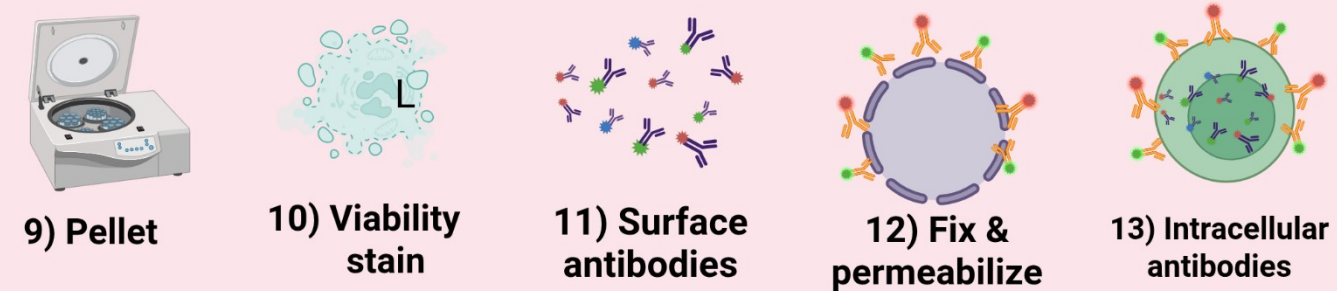
Thaw & Rest



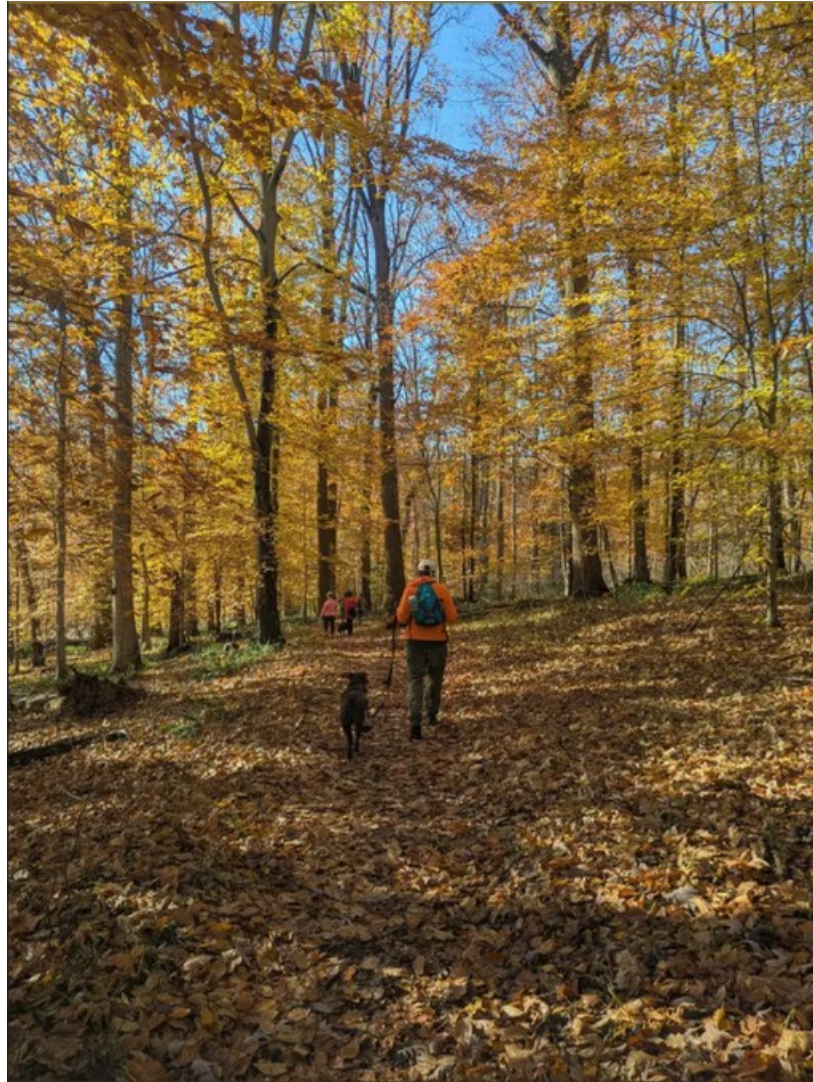
Antigen Incubation



Flow Cytometry Staining



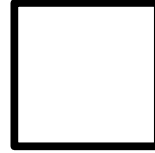
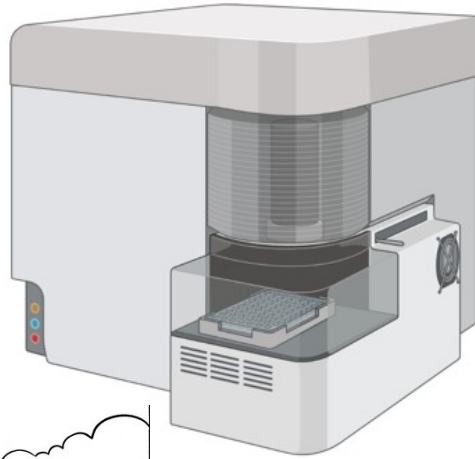
Personal Goal: No cell left behind



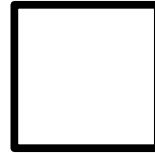
Repeated 3 AM thoughts

- ❑ What makes spectral flow cytometry data amazing is both the depth and the breadth... why in the world would we want to down-sample as a routine step!
- ❑ If acquiring 3 million cells to profile just 100 rare ones, how to fully profile the rest of the dataset to avoid missing other changes

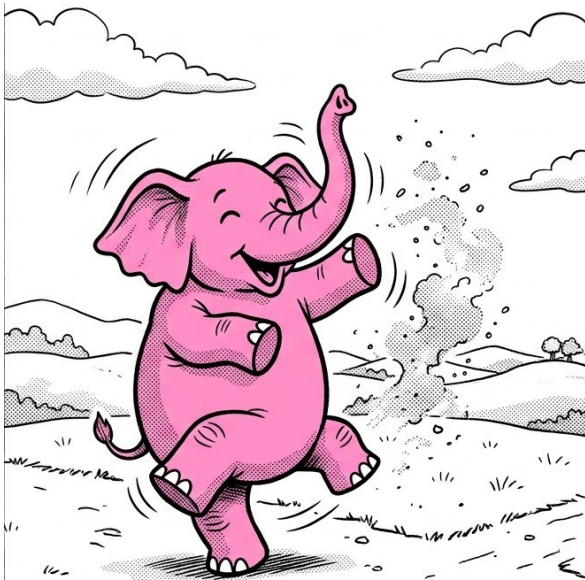
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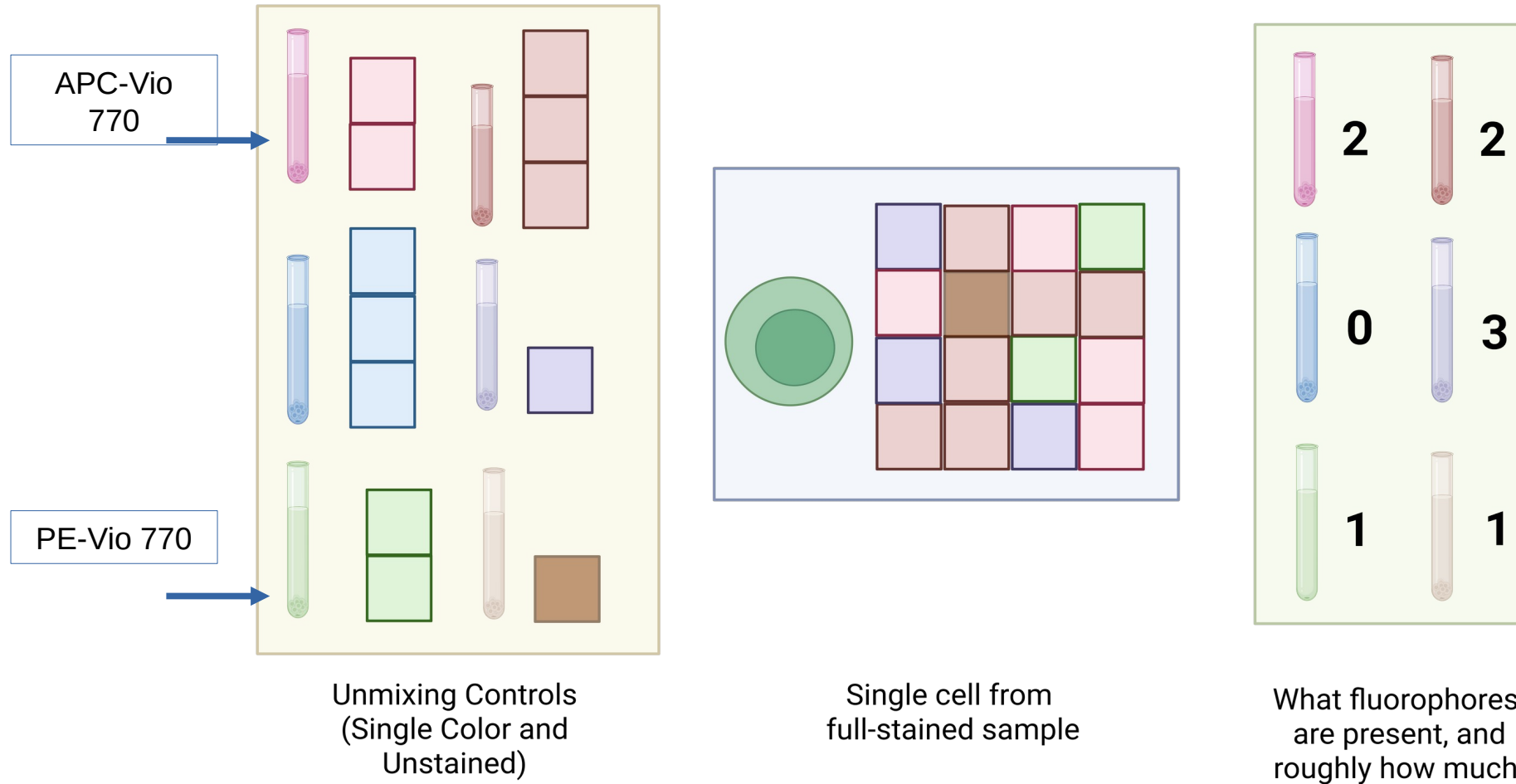
- Quality of the unmixed full-stained data is linked to the quality of our single-color controls (and our ability to provide appropriate autofluorescence controls).



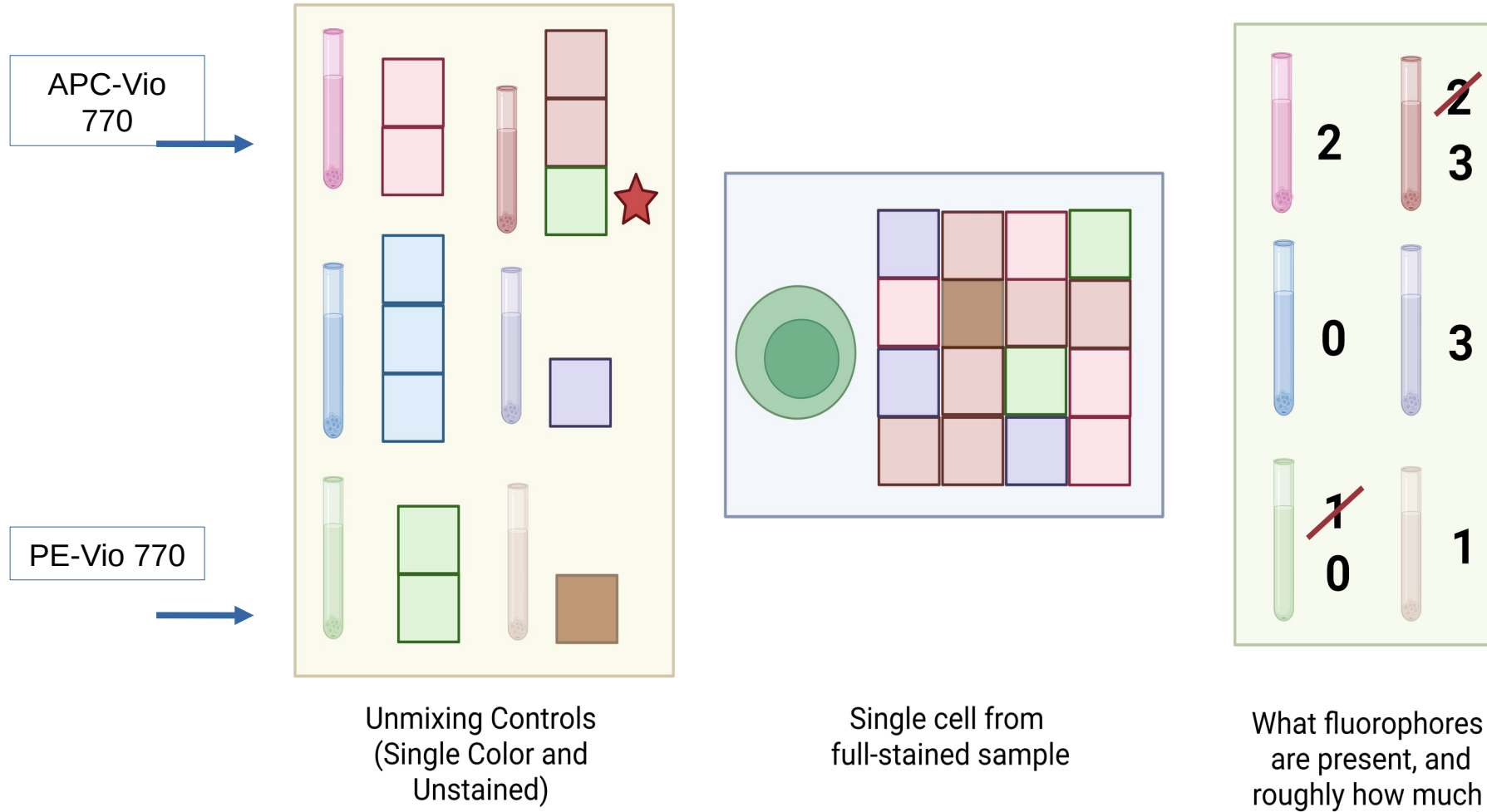
- If we are acquiring 3 million cells to detect 100 rare cells, how do we thoroughly explore the rest of the sample to avoid wasting useful data?



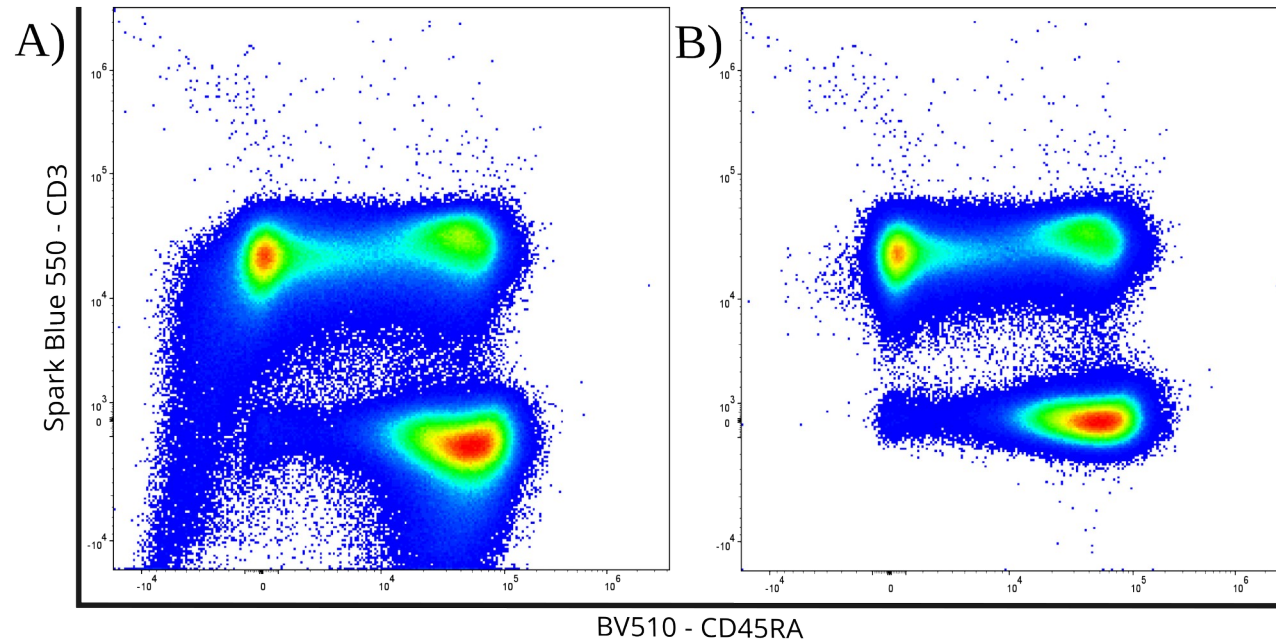
Spectral Unmixing



Mismatched References lead to errors in attribution

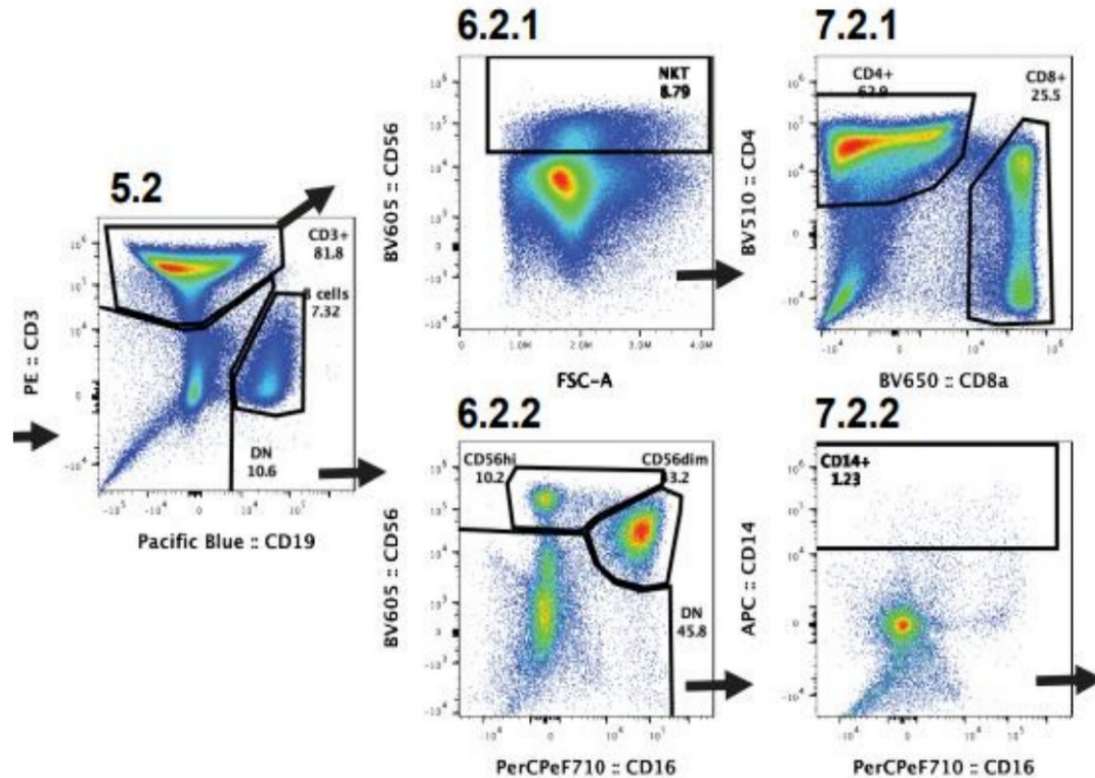


Unmixing Errors



Is the population you are seeing biological, or just a technical artifact?

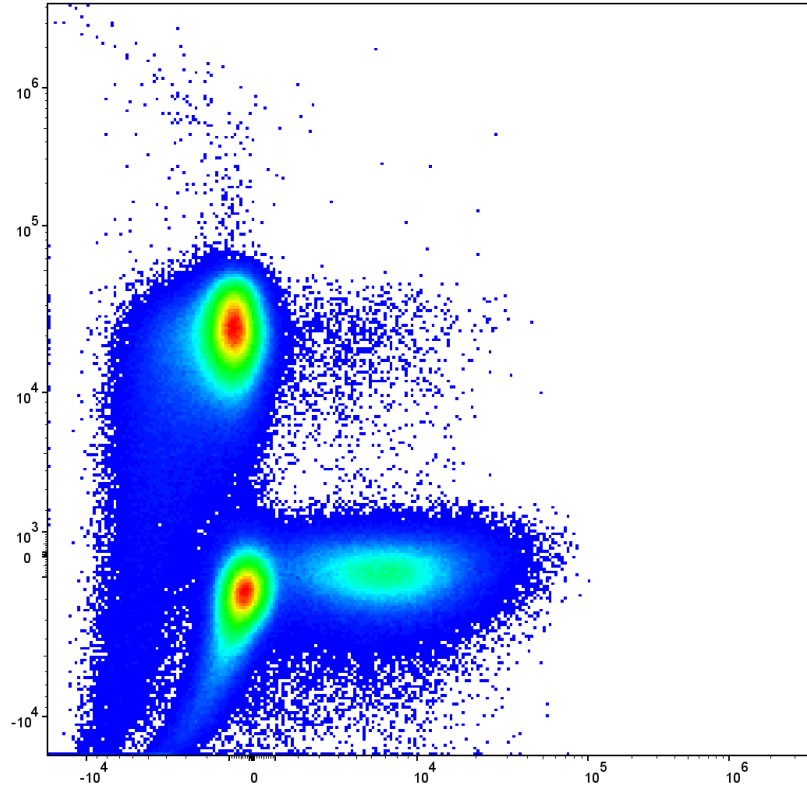
Why does this matter?



* Paper used an identical viability dye signature as the fluorophore of their main marker of interest. No way to tell if their entire paper is based on necrotic cells

- The power to shoot yourself in the foot
- Garbage In, Garbage Out
- Adding to the reproducibility crisis
- Wasting time, effort and funds

Breaking into the unmixing black box



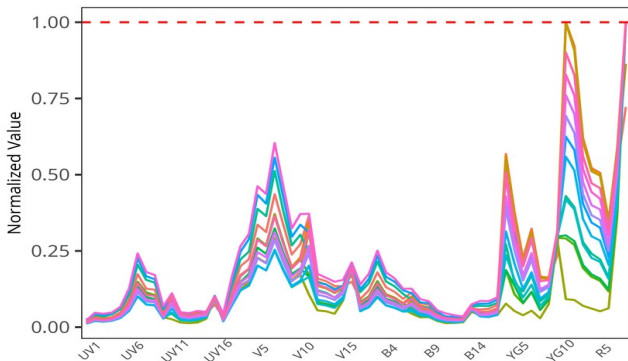
□ Reference signatures extracted from our unmixing controls are critical to the final data quality of the full-stained sample

□ No tools exist to screen the quality of your unmixing control derived signatures pre-unmixing

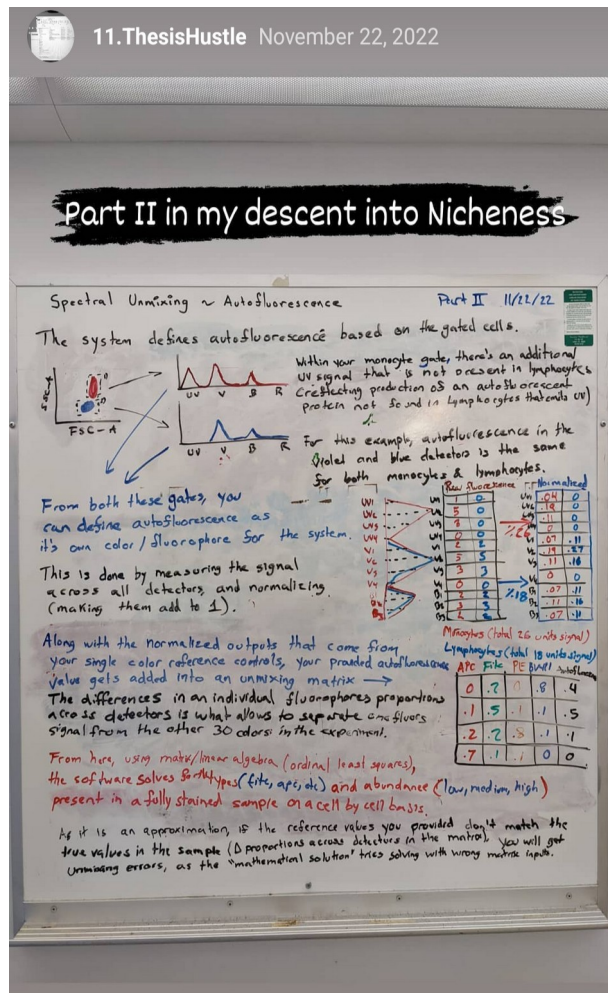
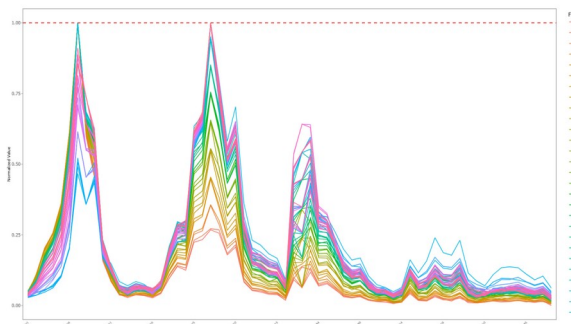
□ Multiple rounds of unmixing trial and error to spot problematic single-color controls

Breaking into the unmixing black box

Tandem Degradation

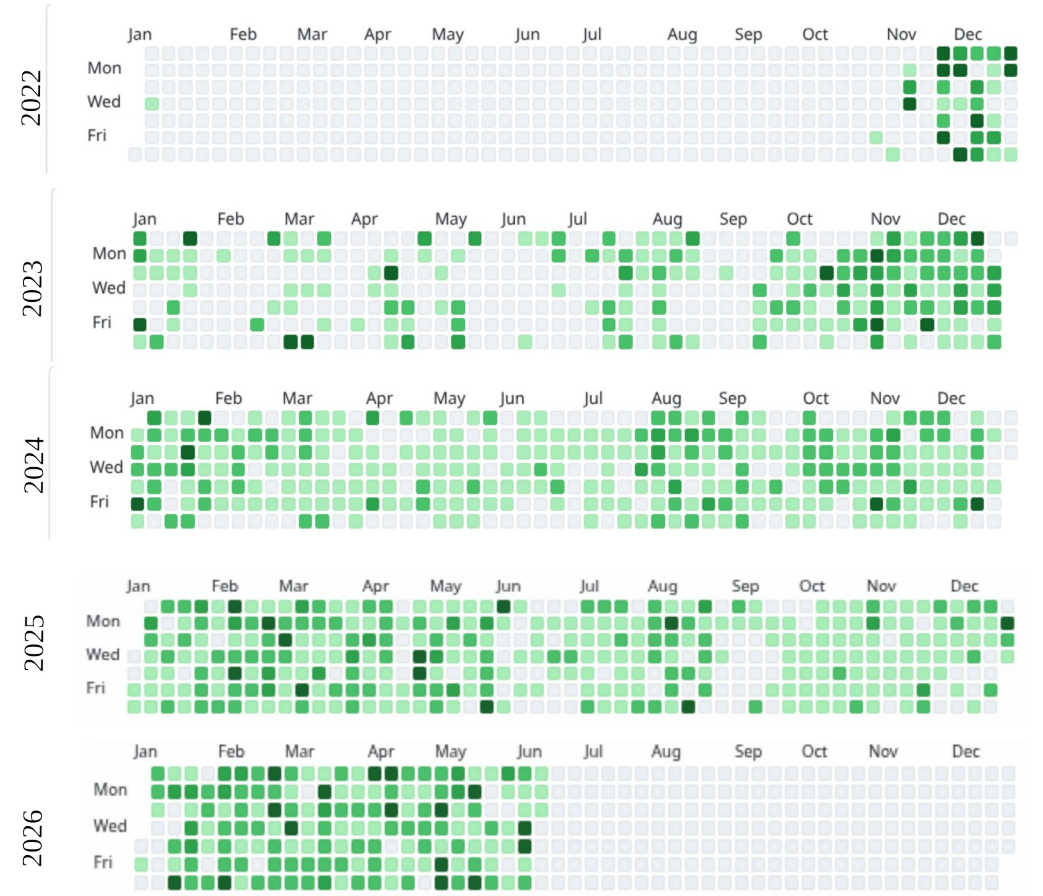
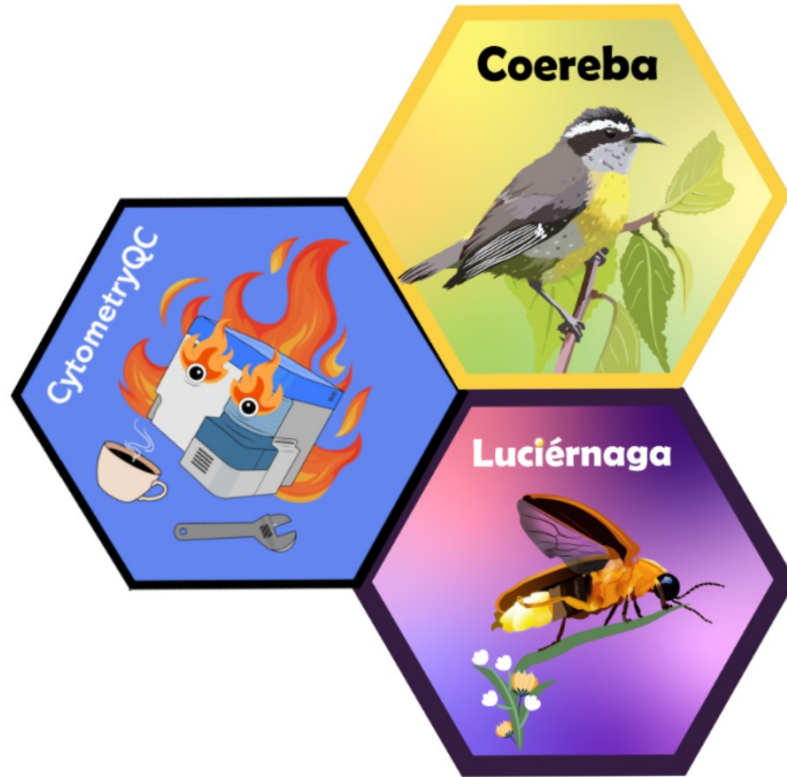


Multiple Autofluorescence



- Reference signatures extracted from our unmixing controls are critical to the final data quality of the full-stained sample
- No tools exist to screen the quality of your unmixing control derived signatures pre-unmixing
- Multiple rounds of unmixing trial and error to spot problematic single-color controls
- What if instead of looking at the signature of the population, we looked at the signatures of the individual cells?

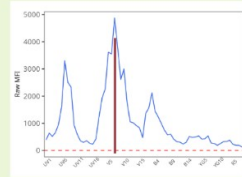
Sparked journey into R programming



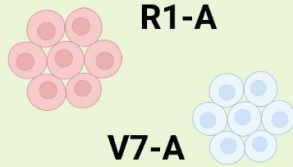
Luciernaga



Code



Identify detector
with brightest
MFI for each cell



Group cells with
shared peak
detectors

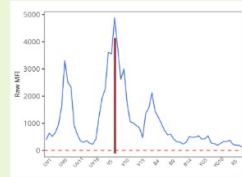


Subtract Out
Autofluorescence

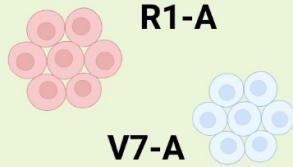
Luciernaga



Code



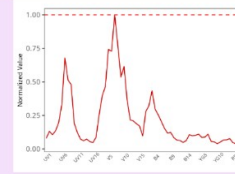
Identify detector
with brightest
MFI for each cell



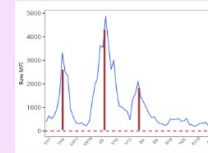
Group cells with
shared peak
detectors



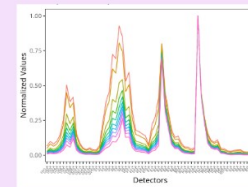
Subtract Out
Autofluorescence



Normalize individual
cell fluorescent
signatures



Identify peaks
using
local maxima

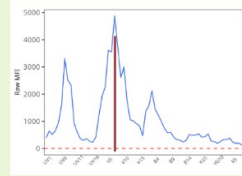


Group cells with
shared peaks &
heights

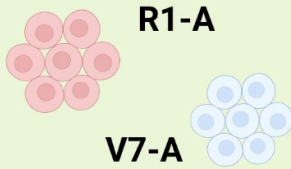
Luciernaga



Code



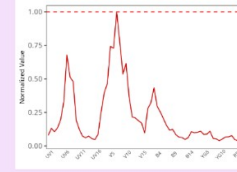
Identify detector
with brightest
MFI for each cell



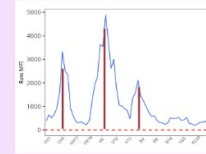
Group cells with shared peak detectors



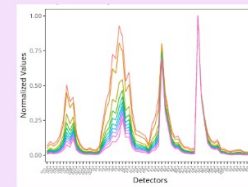
Subtract Out Autofluorescence



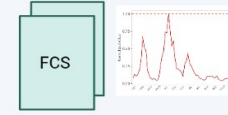
Normalize individual cell fluorescent signatures



Identify peaks
using
local maxima



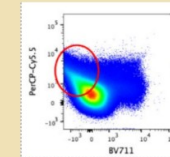
Group cells with
shared peaks &
heights



Enumerate and export out cells that share signature as .fcs files

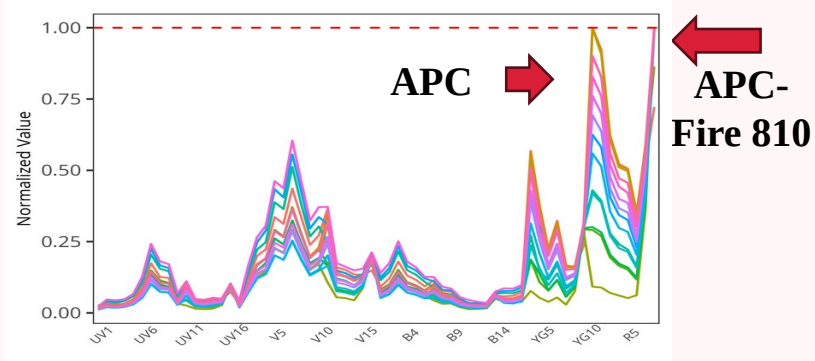
[illegible]

Checks for tandem degradation

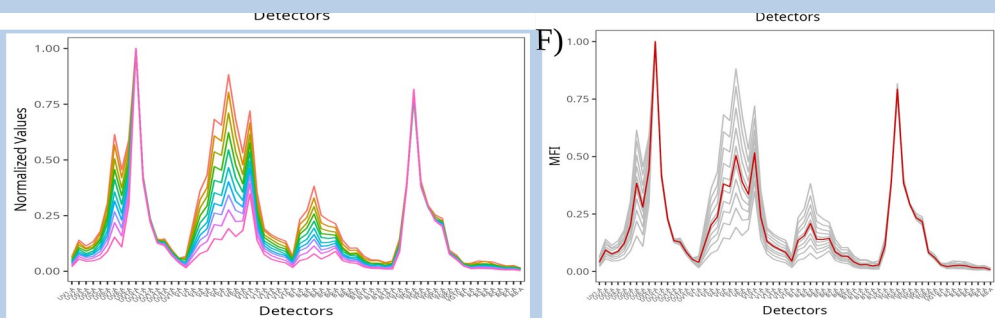


Autofluorescence Isolation

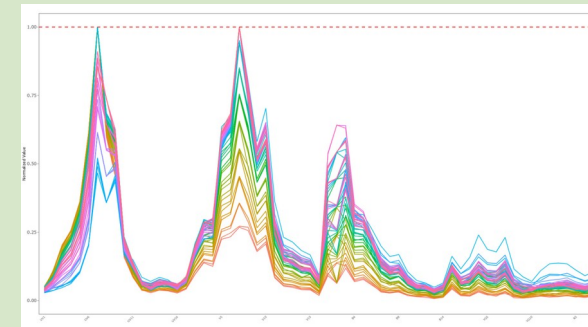
Automated tools for screening unmixing controls



Tandem Degradation



Residual Autofluorescence



Autofluorescence Isolation

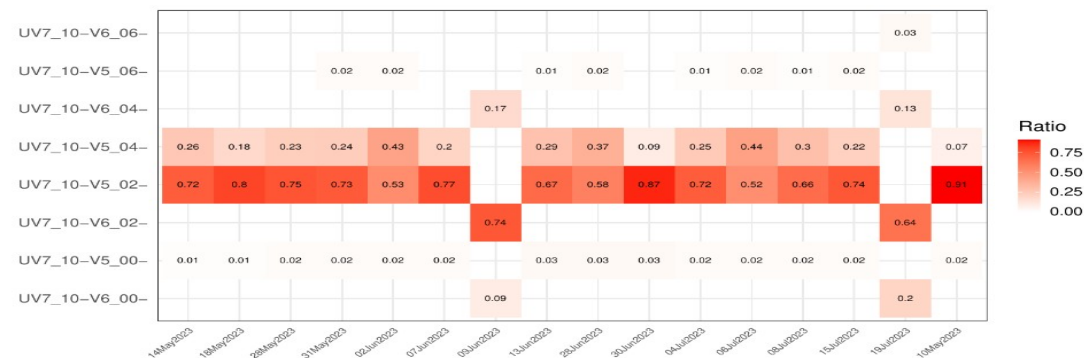
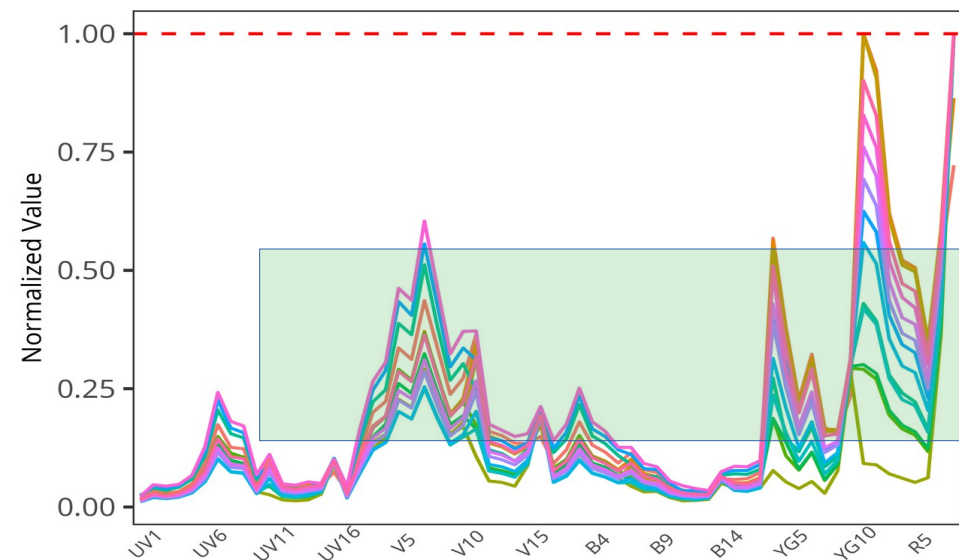
Identifying degrading fluorophores

When looking at normalized signatures of individual cells in the unmixing controls grouped on the basis of shared detectors of relatively same heights

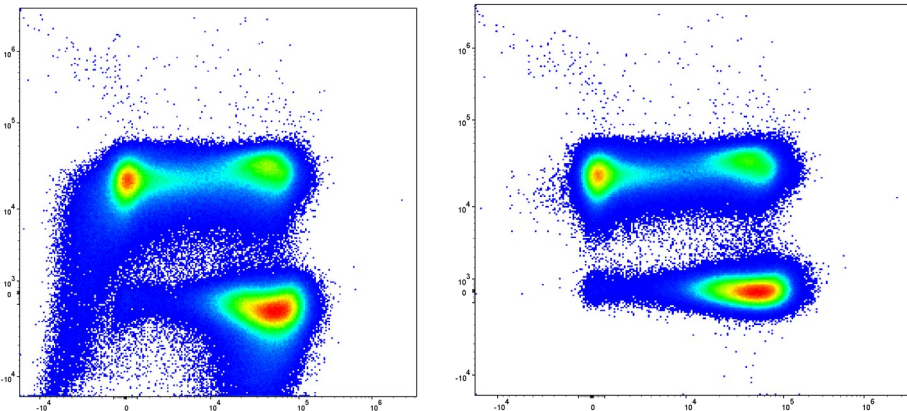
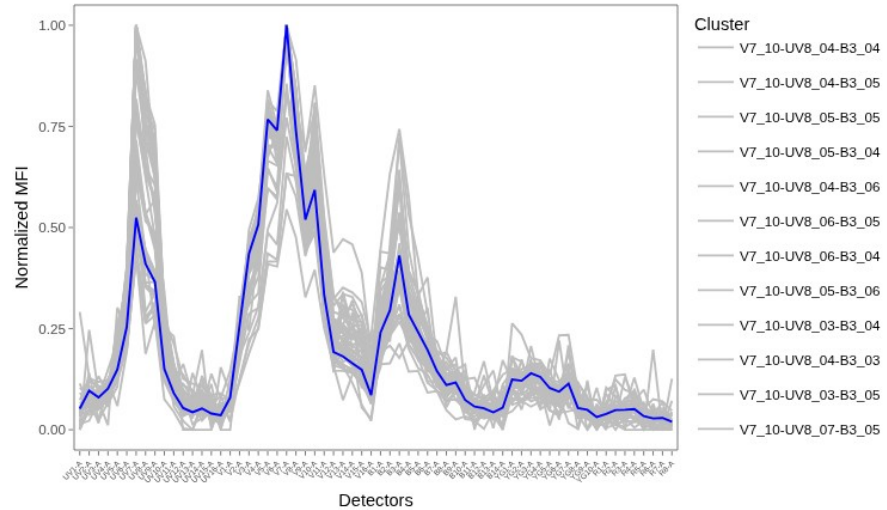
Fluorophores in good shape → **one predominant normalized signature** within the unmixing control raw .fcs file

• When the fluorophore degrades, noticeable **increase in variant signatures**, making up substantial proportion of all cells in the unmixing control

• Automated screening all acquired unmixing controls from campus users during off-hours



Isolating additional autofluorescence signatures

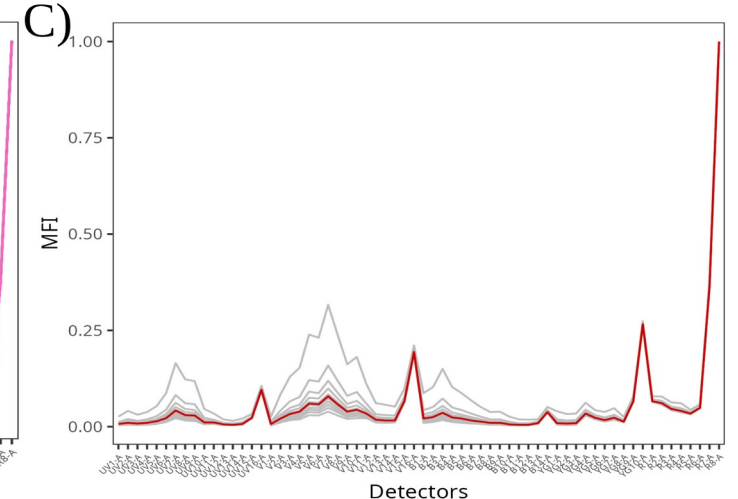
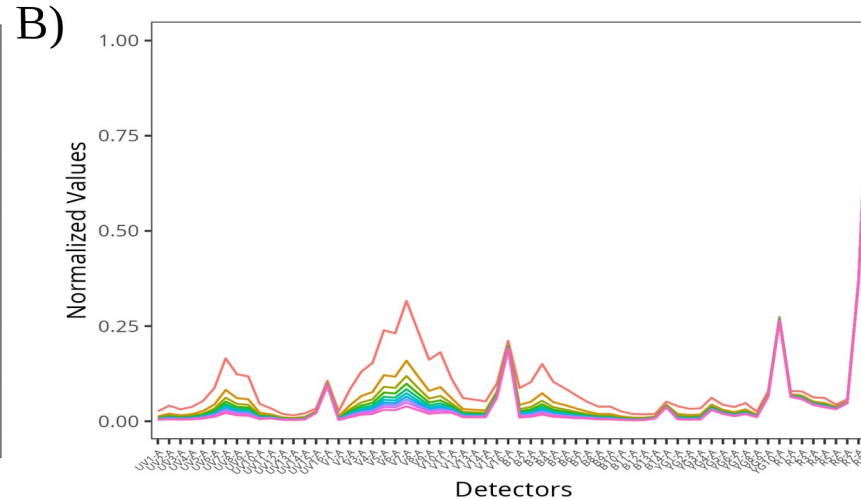
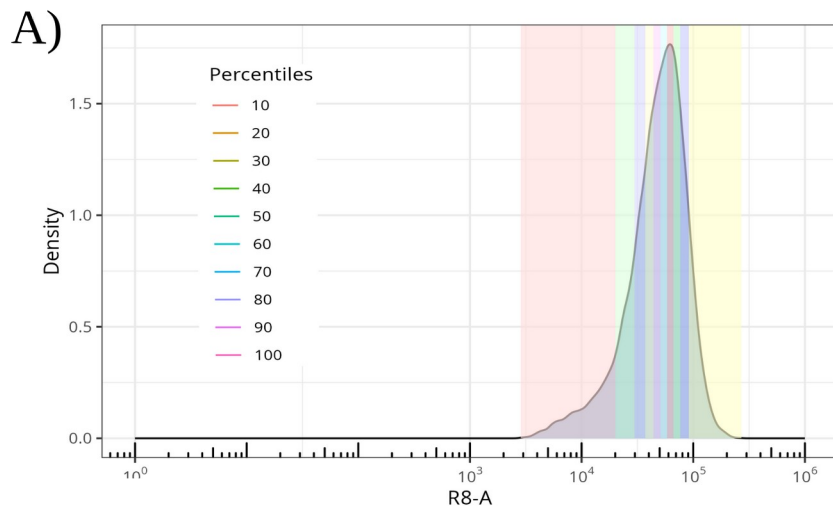


▸ This approach also works to isolate variant autofluorescence signatures from unstained controls

▸ Return directly as an .fcs file containing just cells with same signature for use as an autofluorescent tag in SpectroFlo

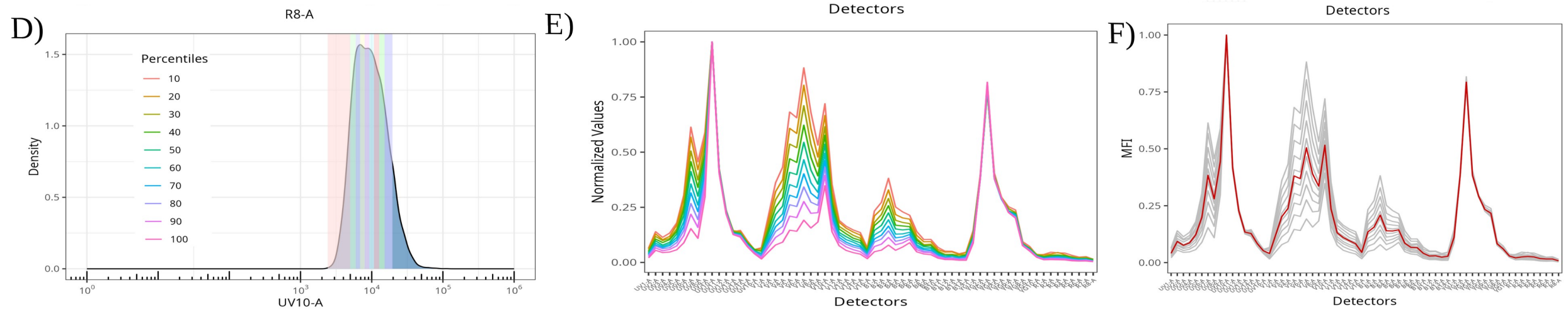
• Unlike other approaches, the isolated cells share normalized signatures, only differing in terms of brightness

PSA: For dim antigen-fluorophore combinations, be cautious using cell unmixing controls



For cell controls with a **moderate density antigen** on **bright fluorophores**, we see that any residual autofluorescence contributes minimally to the overall isolated single-color signature when gated according to best practices

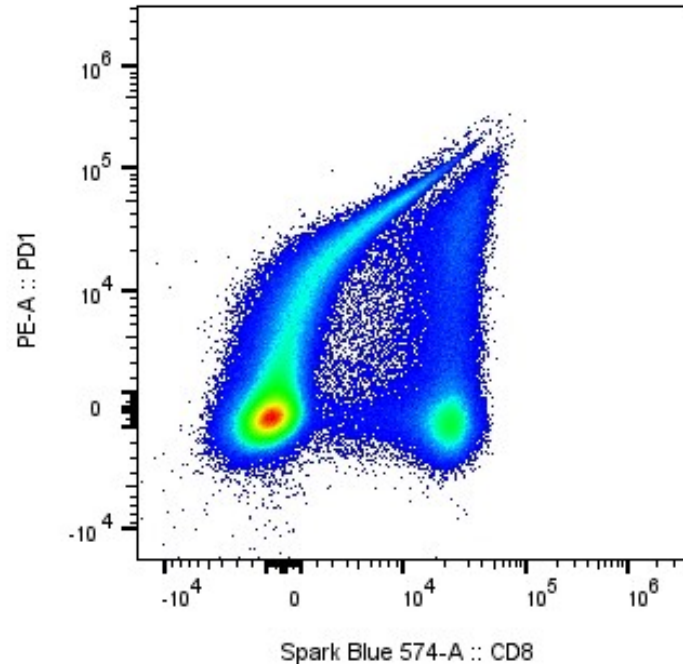
Residual autofluorescence after background subtraction still present in single-color reference signatures



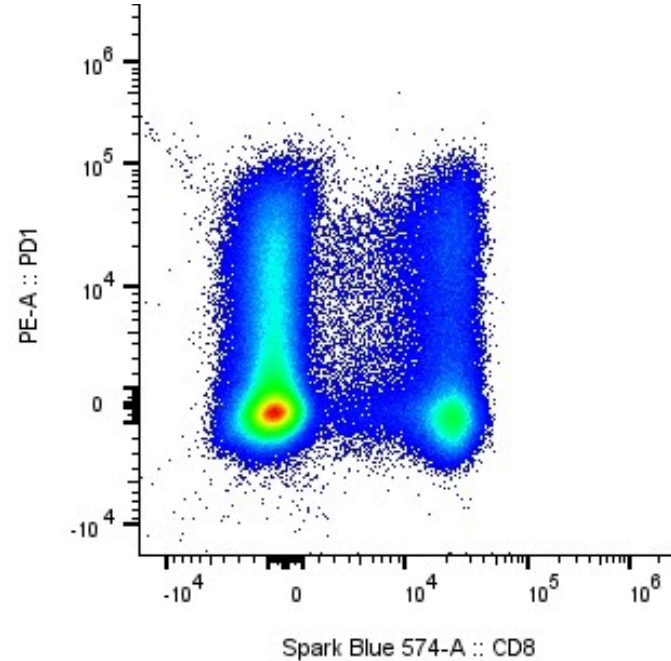
For cell controls with a **moderate density antigen** on **dim fluorophores**, we see substantial residual autofluorescence contributions to the isolated single-color normalized signatures even when gated according to best practices

Swap dim antigen-fluorophore combinations for beads

PE signature with
residual AF

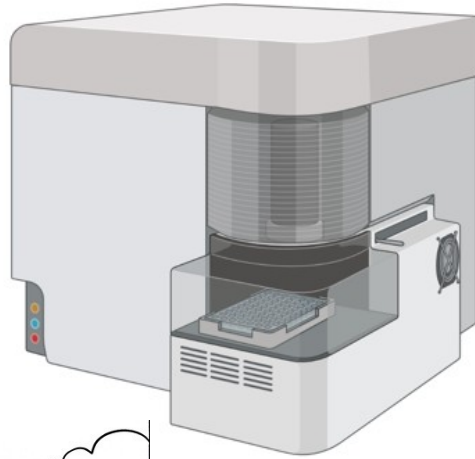


PE signature without
residual AF

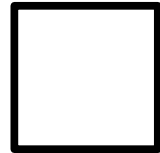


This **residual autofluorescence** contribution **results in unmixing errors**, and when switched out for bead controls the panel unmixing significantly improved

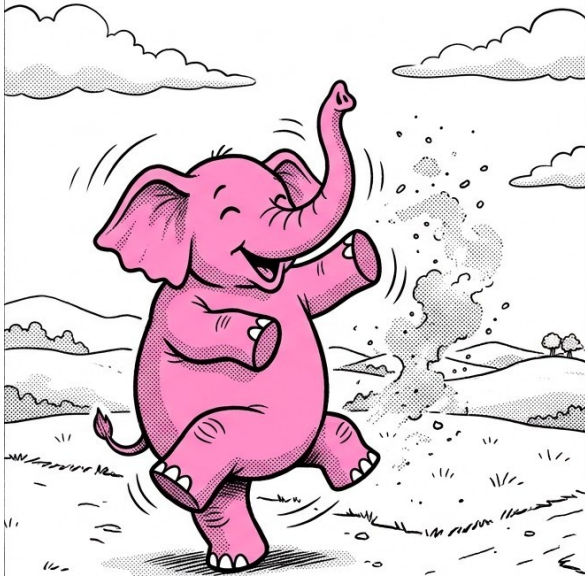
To unlock the comprehensive biological discovery potential, we need to address these pink elephants



- Quality of the unmixed full-stained data is linked to the quality of our single-color controls (and our ability to provide appropriate autofluorescence controls).

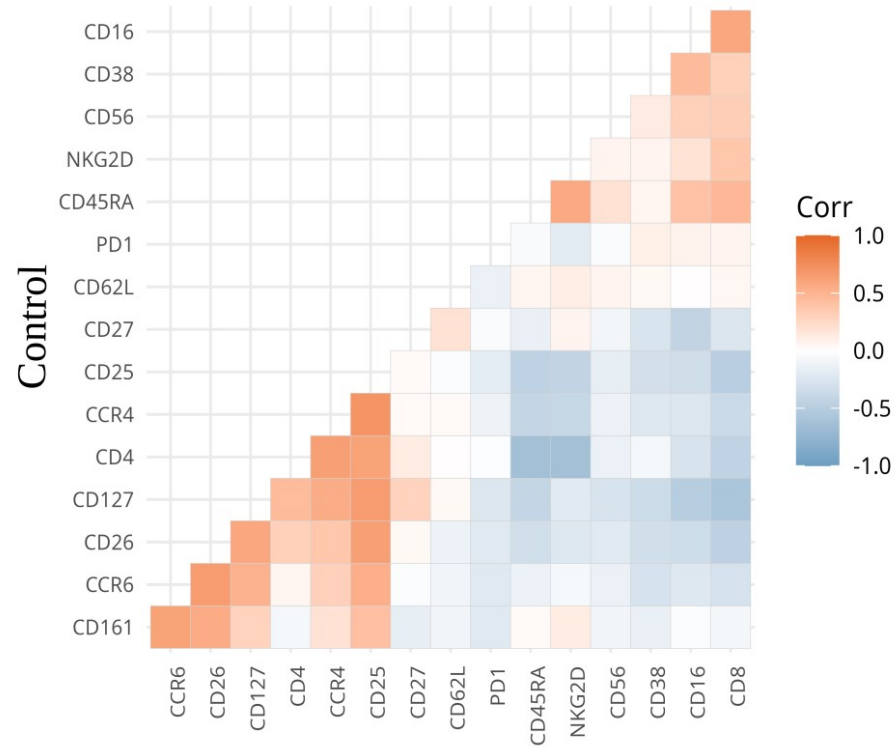


- If we are acquiring 3 million cells to detect 100 rare cells, how do we thoroughly explore the rest of the sample to avoid wasting useful data?



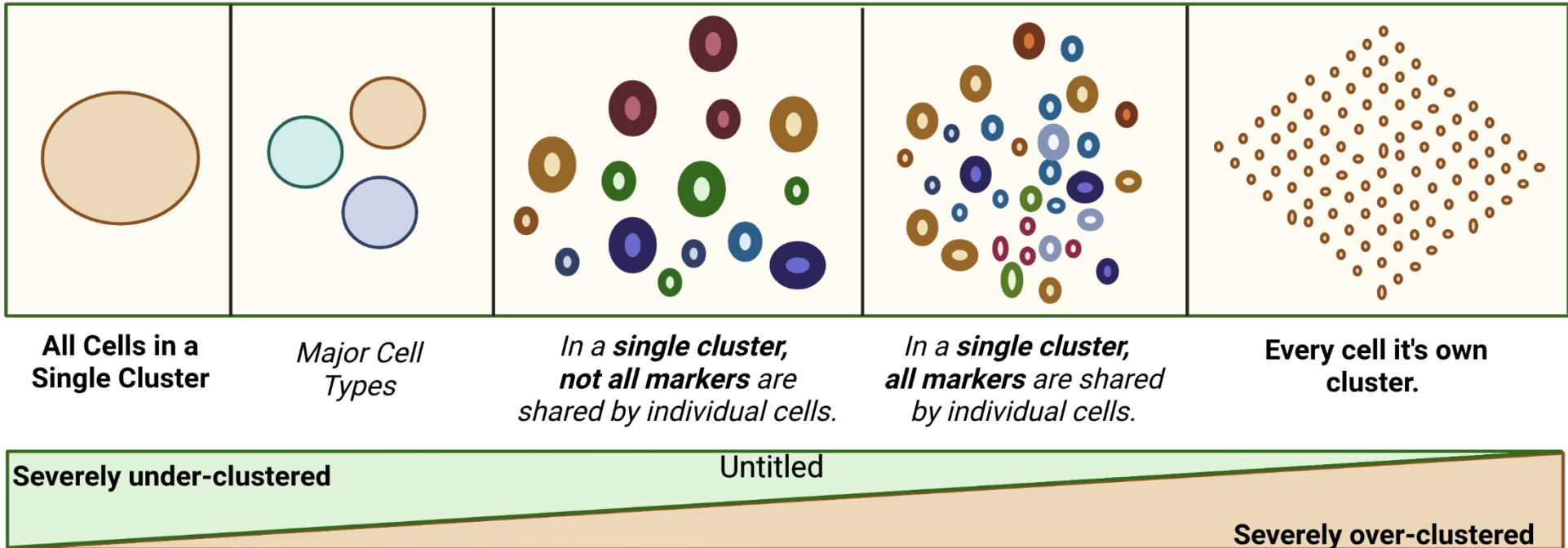
Evidence for additional heterogeneity present in V δ 2

A)

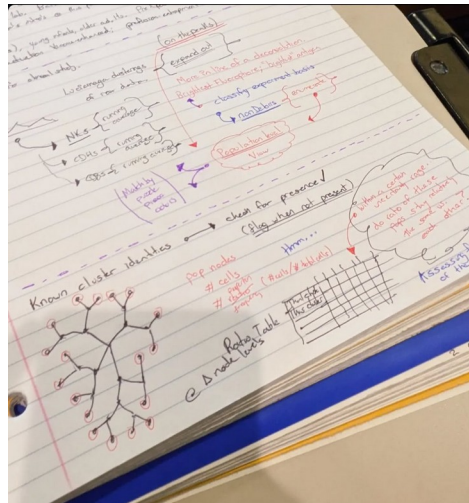
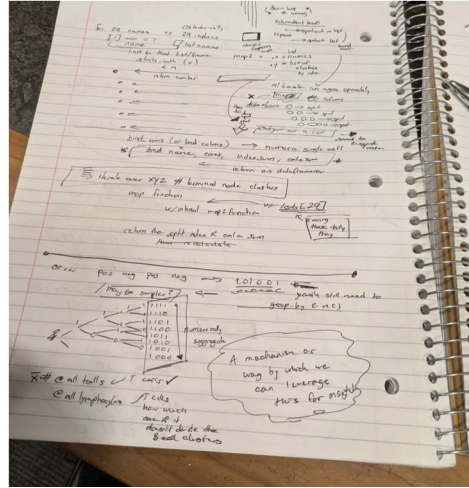
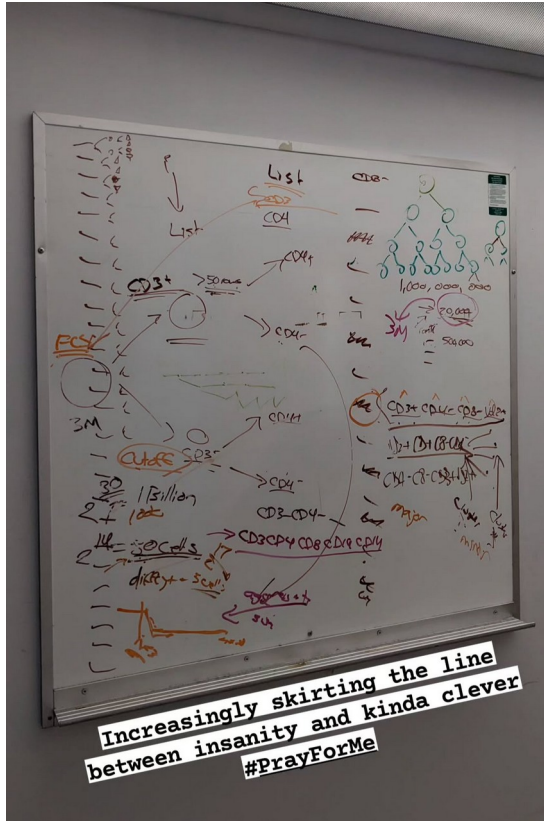


Unlike what we saw for NKTs and MAITs, marker co-expression matrices suggested additional heterogeneity within the V δ 2s

Grouping cells that share every marker



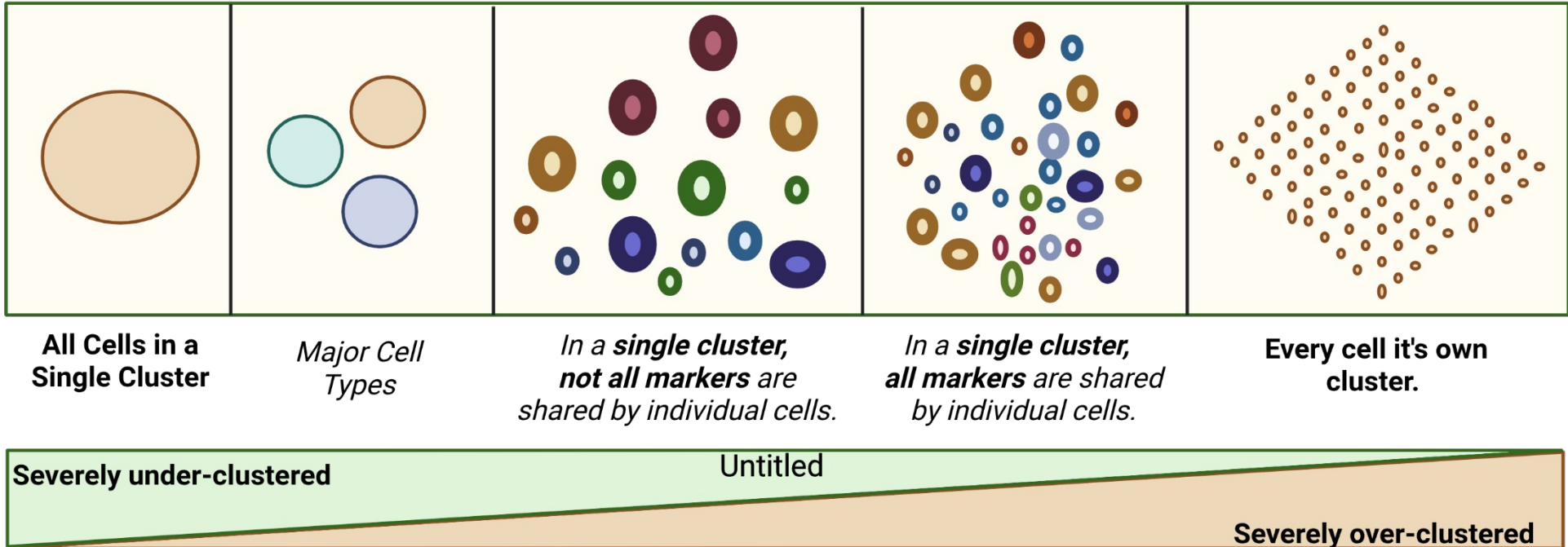
Grouping for shared marker expression



- Different clustering methods tend to inappropriately group or split
- Manual gating doesn't scale
- Automated gates aren't easily adjustable
- Unclear what drives various unsupervised algorithms

→ What if we just annotated manual-gating information in before passing off to the unsupervised algorithms?

Grouping cells that share every marker



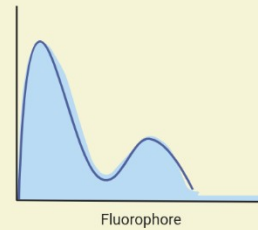
Coereba



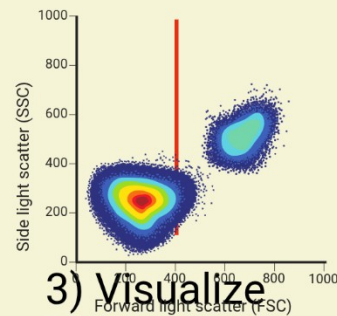
Code



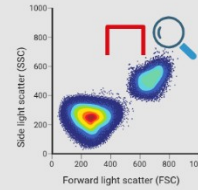
1) Load into R
(flowWorkspace)



2) Calculate
Splitpoints



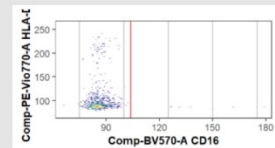
3) Visualize



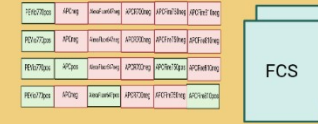
4) Gate Constraints



5) Shiny App
adjustments



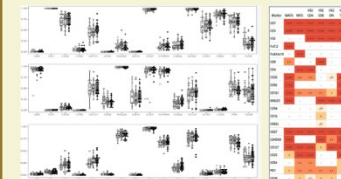
6) Splitpoints
updated



7) CoerebaID



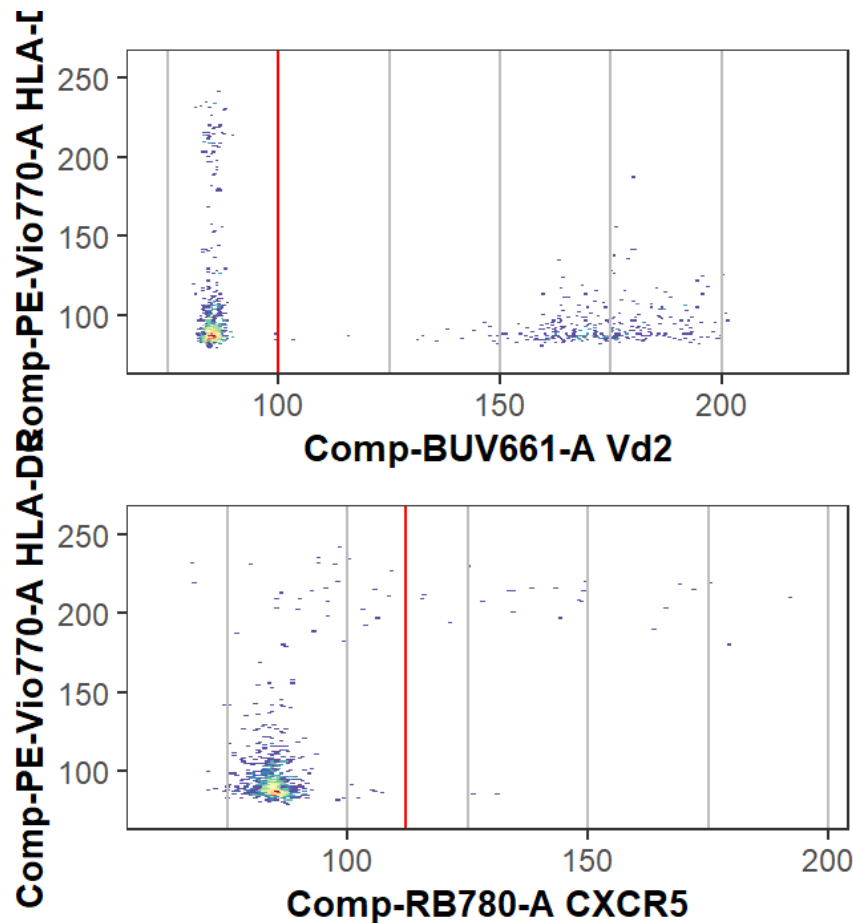
8) Summarized
Experiment



9) Visualize



Initial automated gate, with ability to adjust

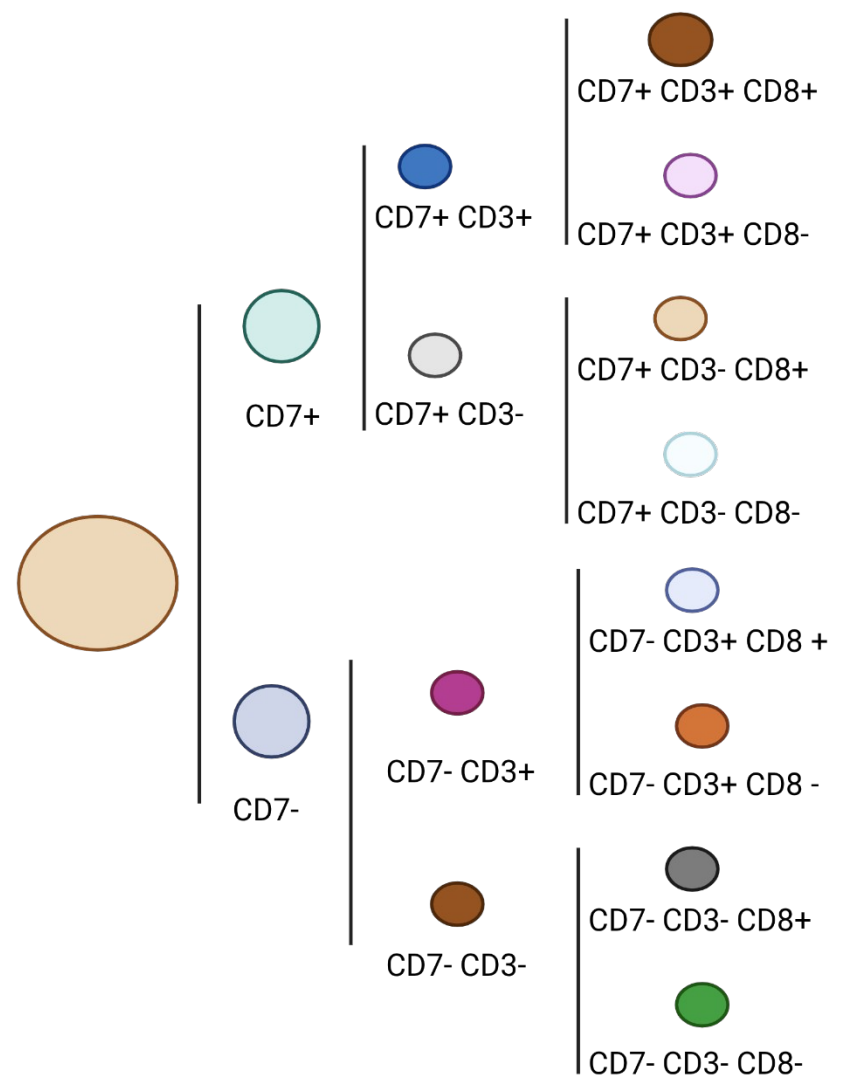


- ☐ Initial gates for designated markers calculated via openCyto (mindensity)
- ☐ Ability to update gate_range constraints
- ☐ Launch a Shiny app and designate interactively

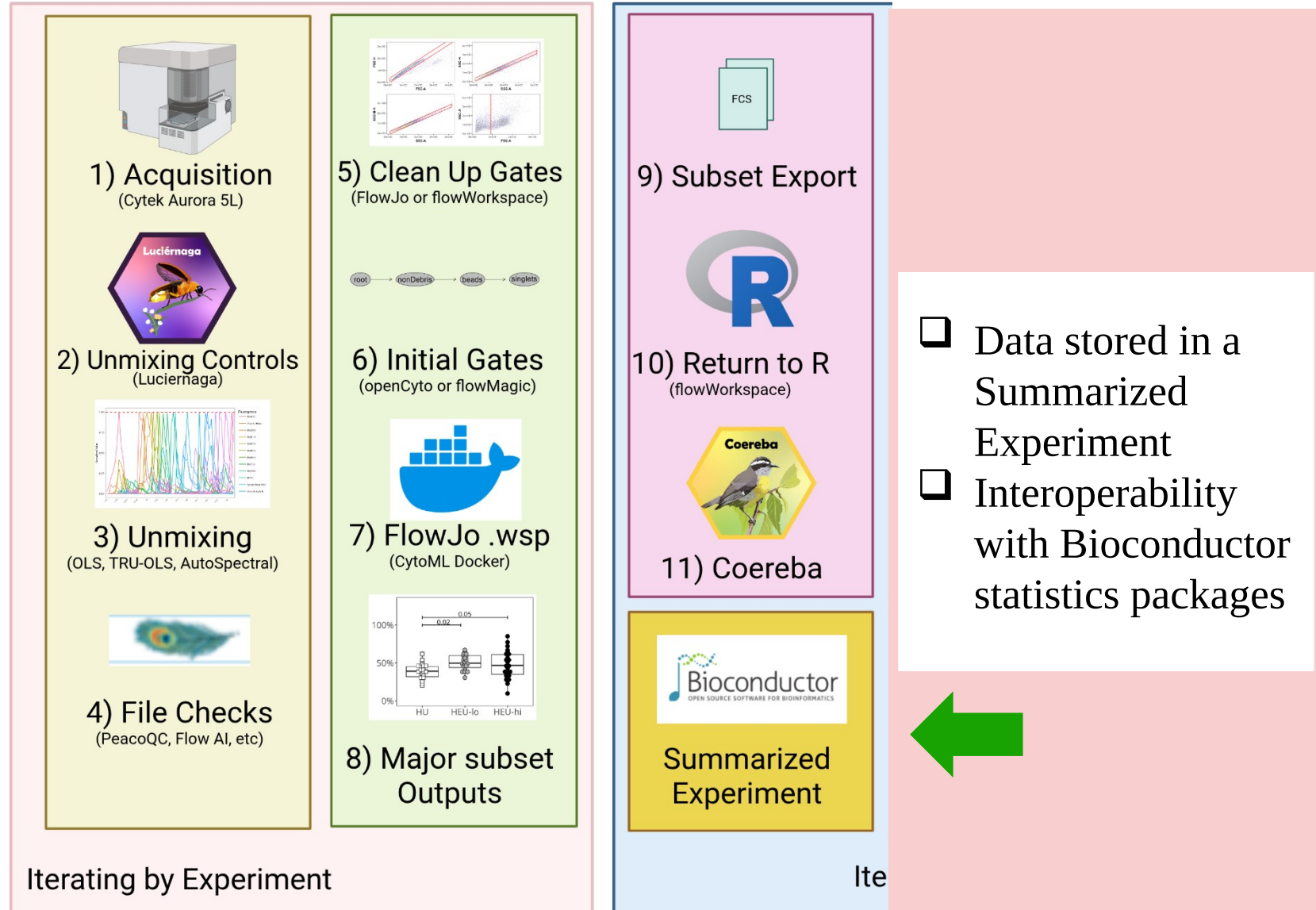


CoerebaID column gets added to the .fcs file

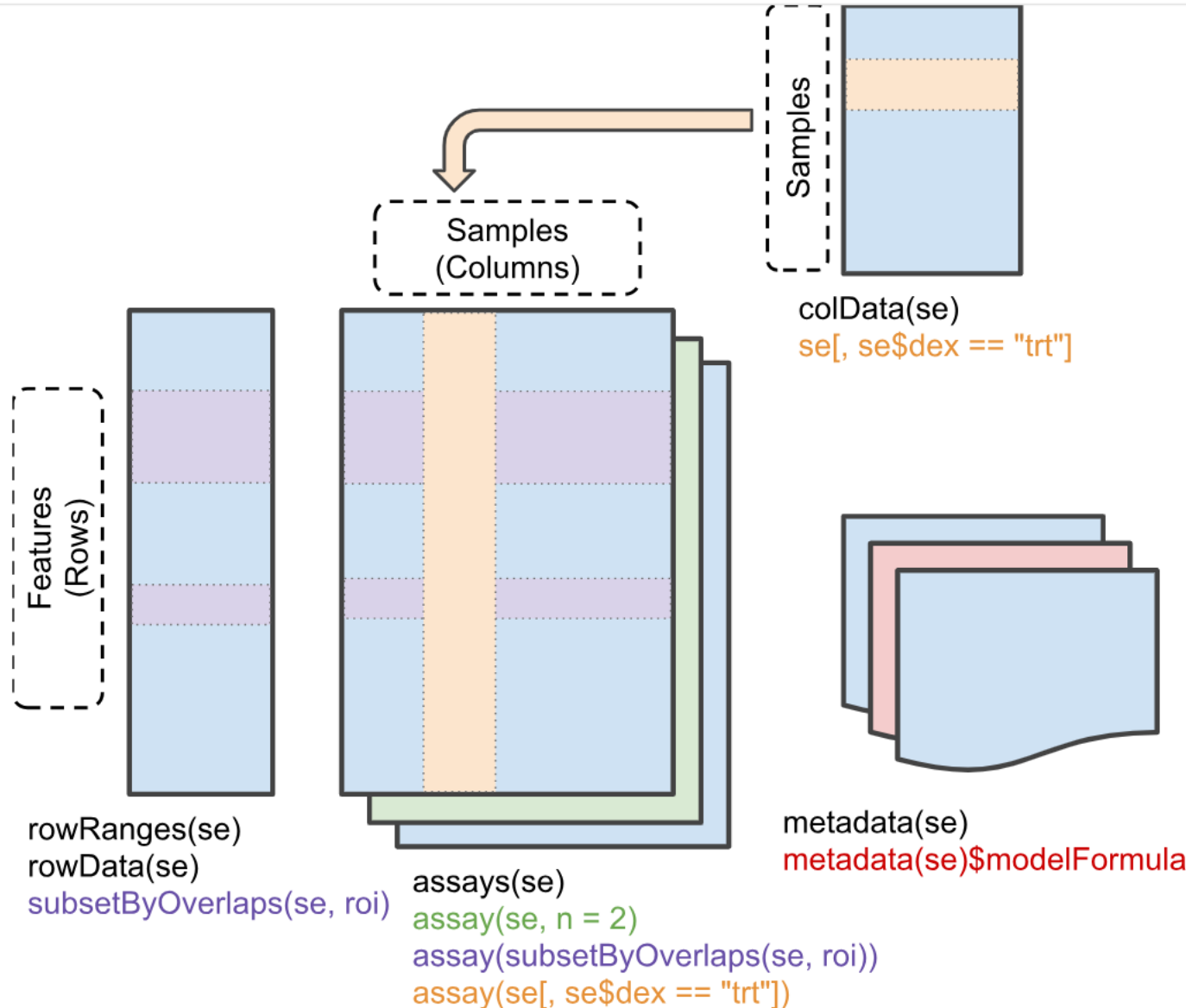
PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCpos	AlexaFluor647neg	APCR700neg	APCFire750pos	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647pos	APCR700neg	APCFire750neg	APCFire810pos



Pipeline: Summarized Experiment



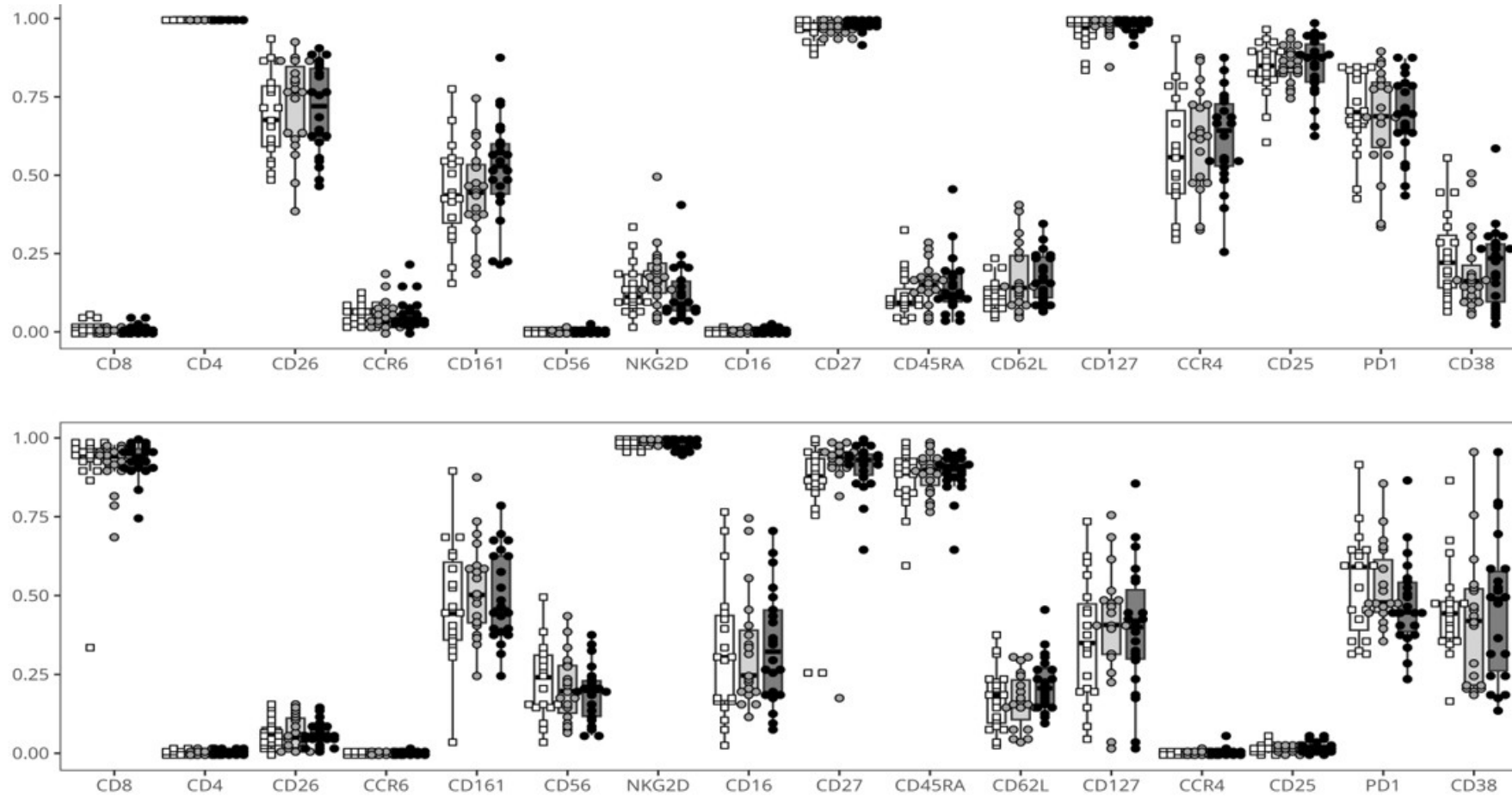
Summarized Experiment



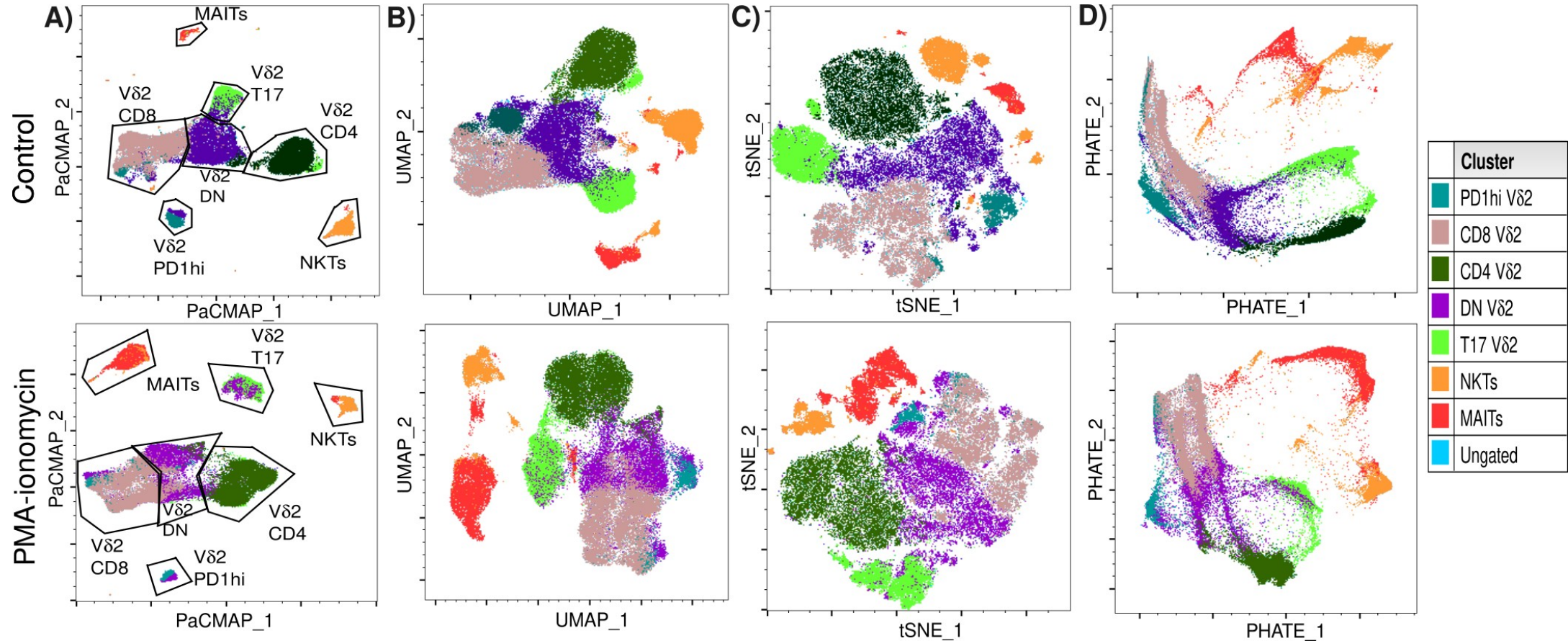
- ❑ Permits rapid re-aggregation of the terminal nodes for re-evaluation
- ❑ Ties in with downstream Bioconductor statistics packages
- ❑ Portable. Don't like how we aggregated groups of cells? File is small enough to store in the same data repository as .fcs files for subsequent re-analysis.
- ❑ Filterable by metadata or node.
 - Nodes only in HEU-hi?
 - Nodes unique to one individual?

Coereba

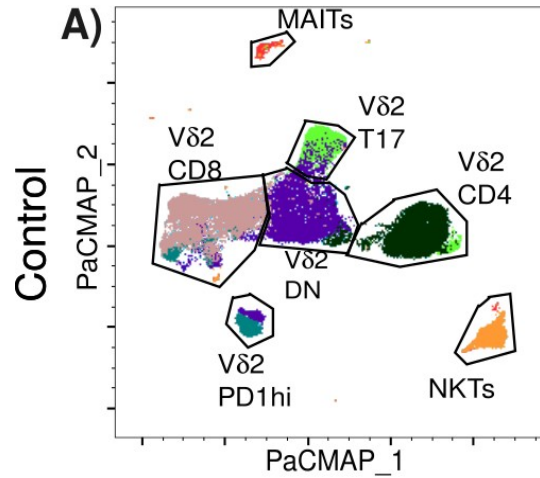
Retrieval of marker expressions



Ability to contrast unsupervised methods vs “manual” ground-truth



Retrieving and re-enumerating

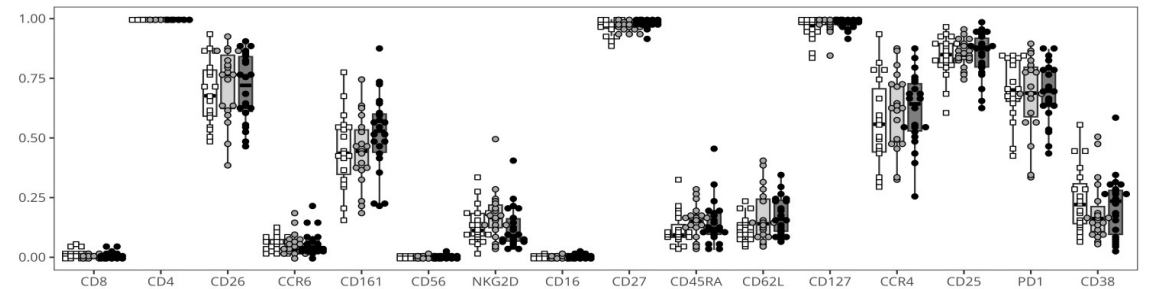


1) Gate

2) Retrieve manual-annotated gates

PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCpos	AlexaFluor647neg	APCR700neg	APCFire750pos	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647pos	APCR700neg	APCFire750neg	APCFire810pos

3) Re-enumerate and plot



Marker expression of cord blood ILT subsets

Control

Marker	MAITs	NKTs	Vδ2 CD4	Vδ2 CD8	Vδ2 DN	Vδ2 T17	Vδ2 PD1
CD7	+++	+++	+++	+++	+++	+++	+++
CD3	+++	+++	+++	+++	+++	+++	+++
Vδ2	-	-	+++	+++	+++	+++	+++
Vα7.2	+++	-	-	-	-	-	-
Vα24Jα18	-	+++	-	-	-	-	-
CD8	++	-	-	+++	-	-	++
CD4	-	+++	+++	-	-	-	-
CD26	+++	++	++	-	-/+	+++	-
CCR6	+++	-	-	-	-	+++	-
CD161	+++	++	++	++	+	+++	-
NKG2D	+++	-	-	+++	+++	+++	-
CD56	-	-	-	-/+	-	-	-/+
CD16	-	-	-	+	-	-	-/+
CD62L	-	-	-	-/+	-	-	-
CD27	+++	+++	+++	+++	+++	+++	+++
CD45RA	+++	-	-	+++	++	+++	+++
CD127	+++	+++	+++	+	+++	+++	-
CD25	+	+++	+++	-	-	+++	-
CCR4	-	++	++	-	-	+	-
PD1	+	++	++	++	++	+	+++
CD38	-	+	-/+	+	+	-	-

Median Frequency
 0-10% 10-30% 30-50% 50-80% 80-100%

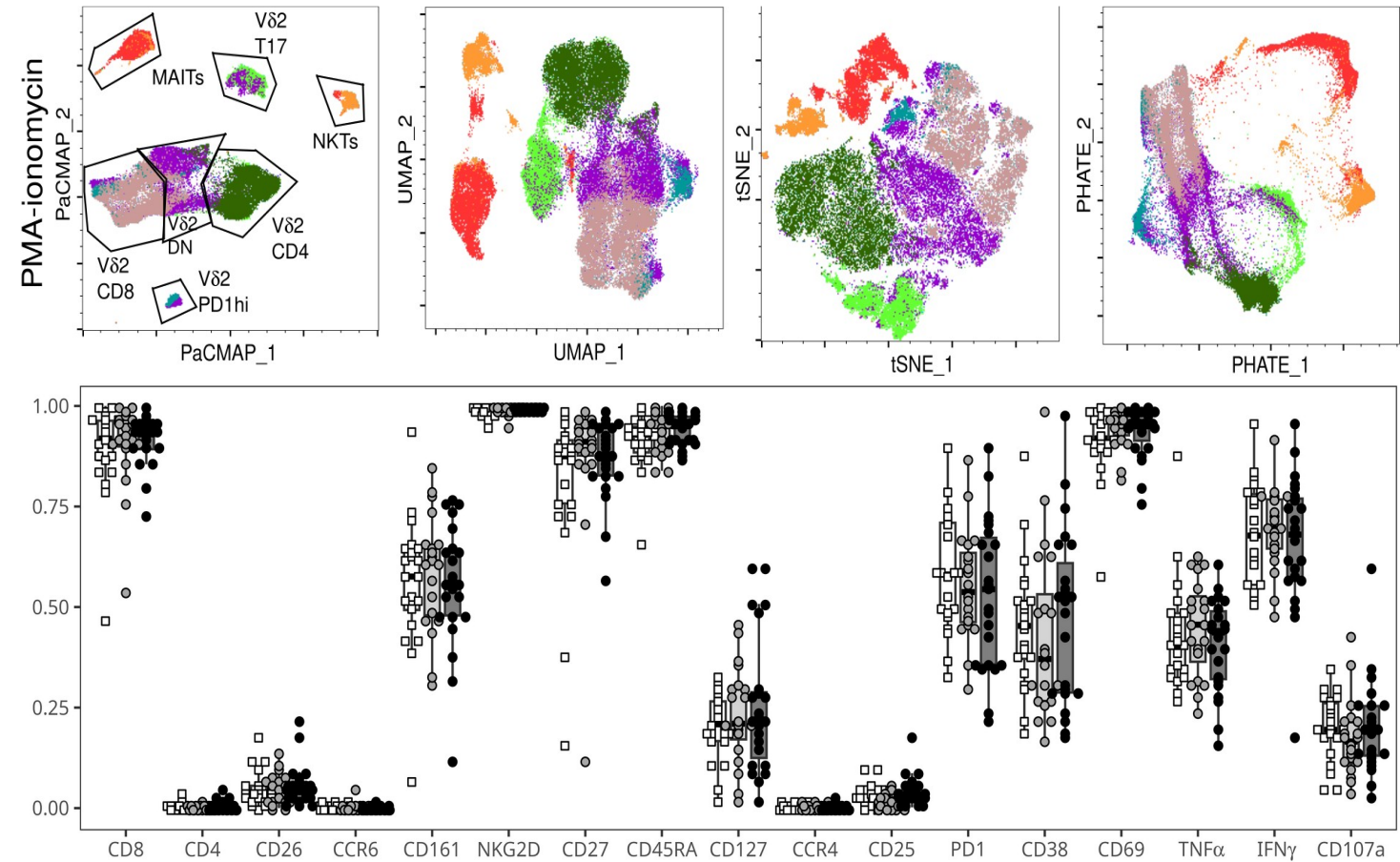
PMA-Ionomycin

Marker	MAITs	NKTs	Vδ2 CD4	Vδ2 CD8	Vδ2 DN	Vδ2 T17	Vδ2 PD1
CD7	+++	+++	+++	+++	+++	+++	+++
CD3	+++	+++	+++	+++	+++	+++	+++
Vδ2	-	-	+++	+++	+++	+++	+++
Vα7.2	+++	-	-	-	-	-	-
Vα24Jα18	-	+++	-	-	-	-	-
CD8	++	-	-	+++	-	-	++
CD4	-	+++	+++	-	-	-	-
CD26	+++	++	++	-	-/+	+++	-
CCR6	++	-	-	-	-	+++	-
CD161	+++	++	++	++	+	+++	-
NKG2D	+++	-	-	+++	+++	+++	-
CD56	-	-	-	-	-	-	-
CD16	-	-	-	-	-	-	-
CD62L	-	-	-	-	-	-	-
CD27	+++	+++	+++	+++	+++	+++	+++
CD45RA	+++	-	-	+++	++	++	+++
CD127	+++	+++	+++	-/+	++	+++	-
CD25	+	+++	+++	-	-	+++	-
CCR4	-	++	-/+	-	-	-	-
PD1	++	+++	+++	++	+++	++	+++
CD38	-	++	-/+	+	+	-	-
CD69	+++	+++	+++	+++	+++	+++	++
TNFα	+	+++	+++	+	+	+	-
IFNγ	-	-/+	++	++	+	-	-/+
CD107a	-	-	-	-/+	-	-	-/+

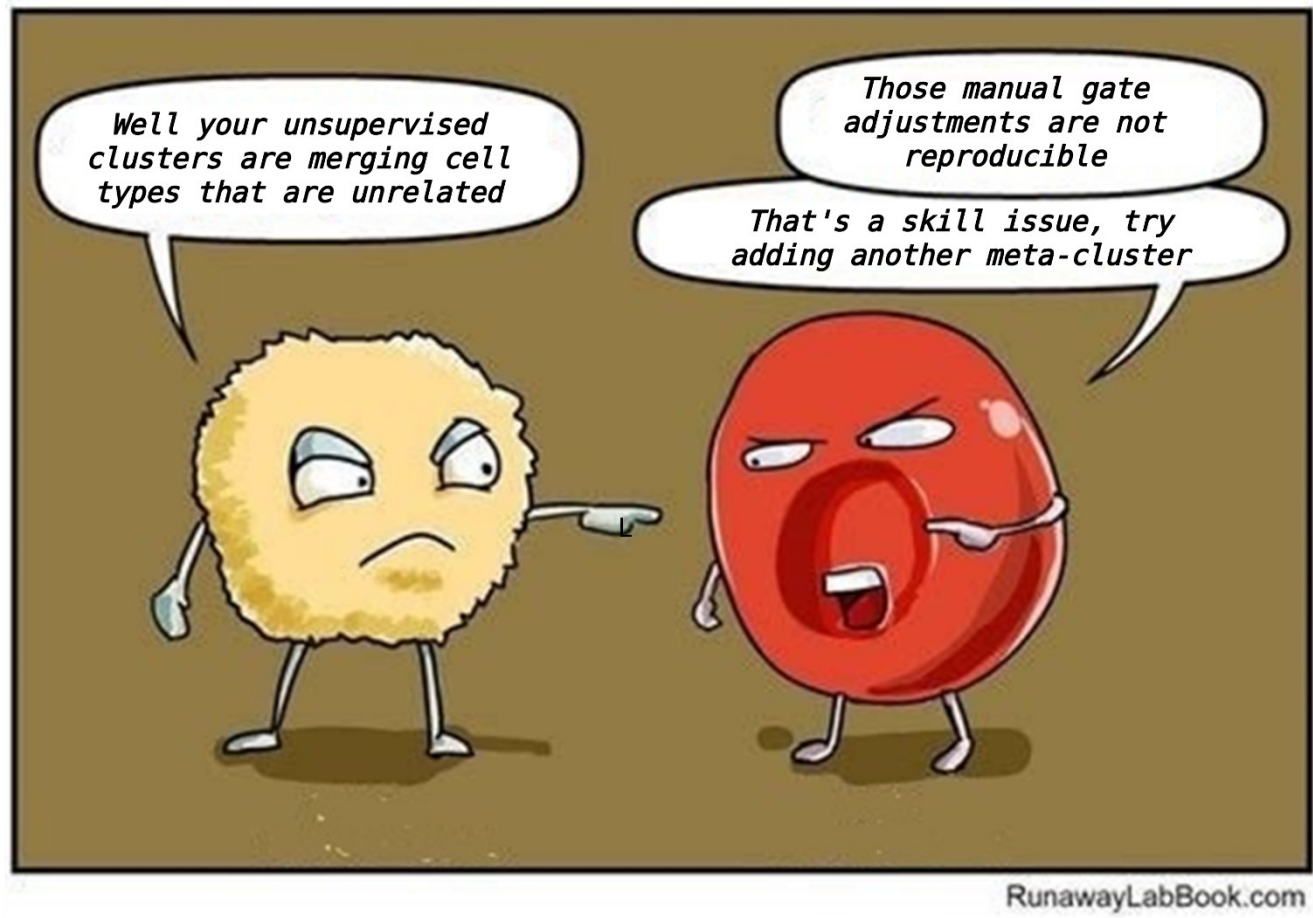
Median Frequency
 0-10% 10-30% 30-50% 50-80% 80-100%

Retrieve manual annotated gating information following unsupervised steps

Marker	MAITs	NKTs	Vδ2 CD4	Vδ2 CD8	Vδ2 DN	Vδ2 T17	Vδ2 PD1
CD7	+++	+++	+++	+++	+++	+++	+++
CD3	+++	+++	+++	+++	+++	+++	+++
Vδ2	-	-	+++	+++	+++	+++	+++
Vα7.2	+++	-	-	-	-	-	-
Vα24Ja18	-	+++	-	-	-	-	-
CD8	++	-	-	+++	-	-	++
CD4	-	+++	+++	-	-	-	-
CD26	+++	++	++	-	-/+	+++	-
CCR6	+++	-	-	-	-	+++	-
CD161	+++	++	++	++	+	+++	-
NKG2D	+++	-	-	+++	+++	+++	-
CD56	-	-	-	-/+	-	-	-/+
CD16	-	-	-	+	-	-	-/+
CD62L	-	-	-	-/+	-	-	-
CD27	+++	+++	+++	+++	+++	+++	+++
CD45RA	+++	-	-	+++	++	+++	+++
CD127	+++	+++	+++	+	+++	+++	-
CD25	+	+++	+++	-	-	+++	-
CCR4	-	++	++	-	-	+	-
PD1	+	++	++	++	++	+	+++
CD38	-	+	-/+	+	+	-	-



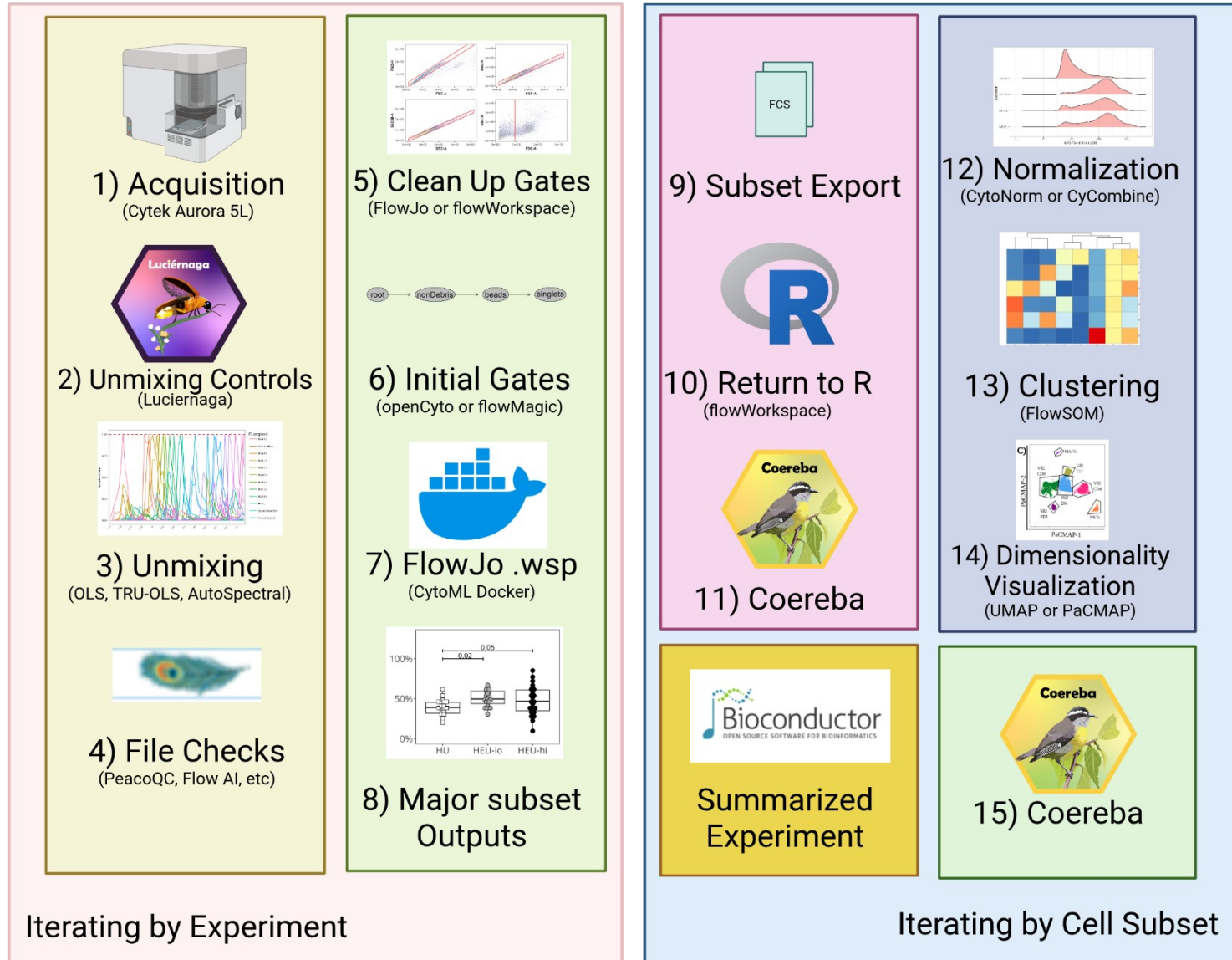
Functional and Iterative: plenty for everyone to dislike :)



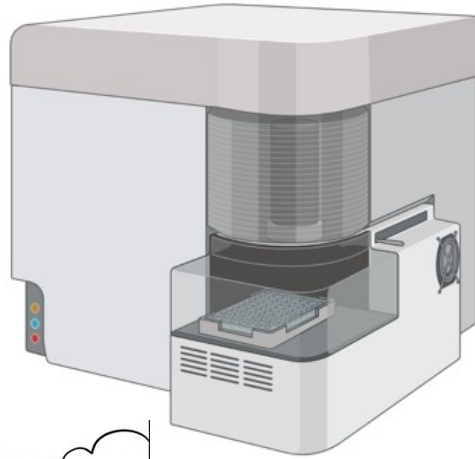
Fun off-label uses thus far:

- ❑ Contrasting where “manually-gated” cells end up vs. our unsupervised methods (normalization, clustering, etc.)
- ❑ Added benefit of looking-at-the-forest: Easy to identify badly cryopreserved specimens where cell populations were lost

Semi-supervised Pipeline



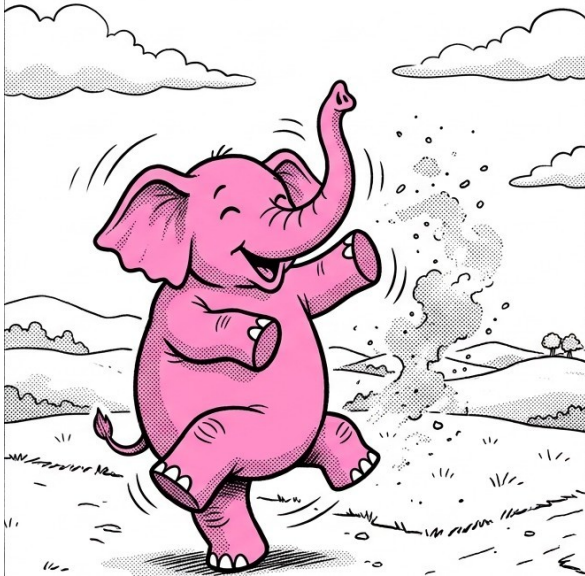
To unlock the comprehensive biological discovery potential, we need to address these pink elephants



- Quality of the unmixed full-stained data is linked to the quality of our single-color controls (and our ability to provide appropriate autofluorescence controls).



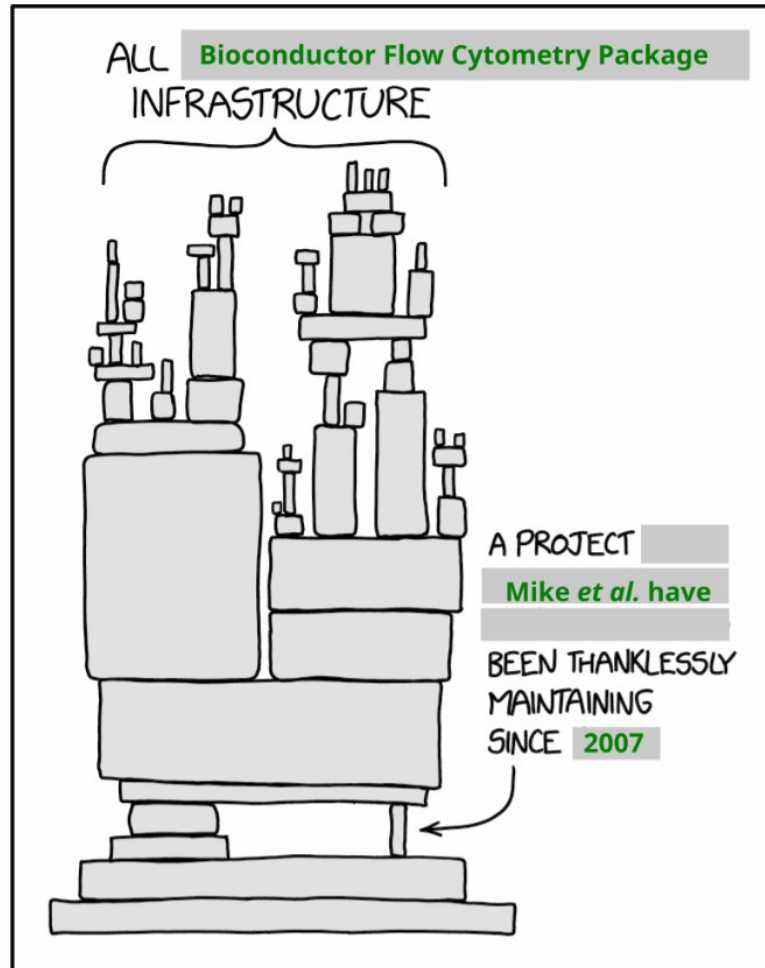
- If we are acquiring 3 million cells to detect 100 rare cells, how do we thoroughly explore the rest of the sample to avoid wasting useful data?



So are all our problems solved?



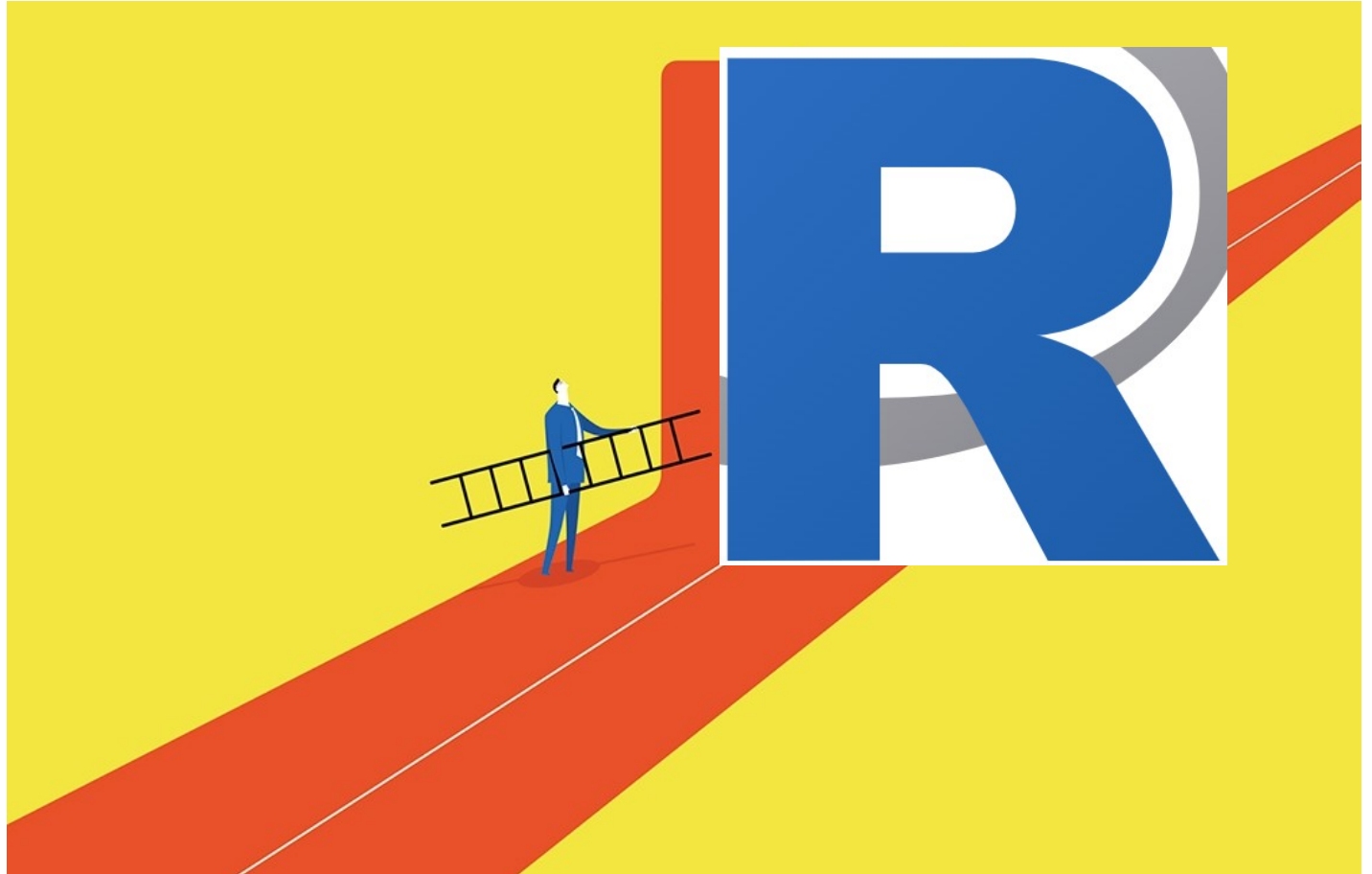
What good is any tool if the community is not able to use it?



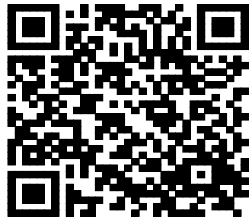
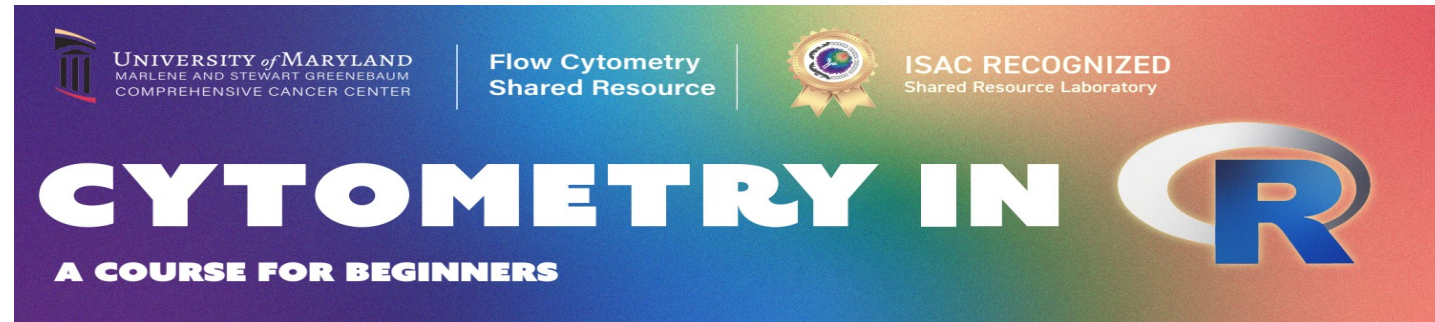
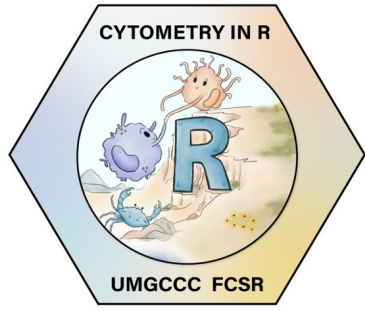
- Working towards formally submitting my R packages to Bioconductor this year (adding documentation, generalizing for different instrument manufacturers)
 - Still learning better ways to parallelize and speed up the process
-
- 72 Bioconductor cytometry packages (+ 20? GitHub based packages)
 - Heavily under-utilized, some unmaintained
 - Evidence that “Build the `software tool` and they will come” approach doesn’t work in practice

Major barrier to entry

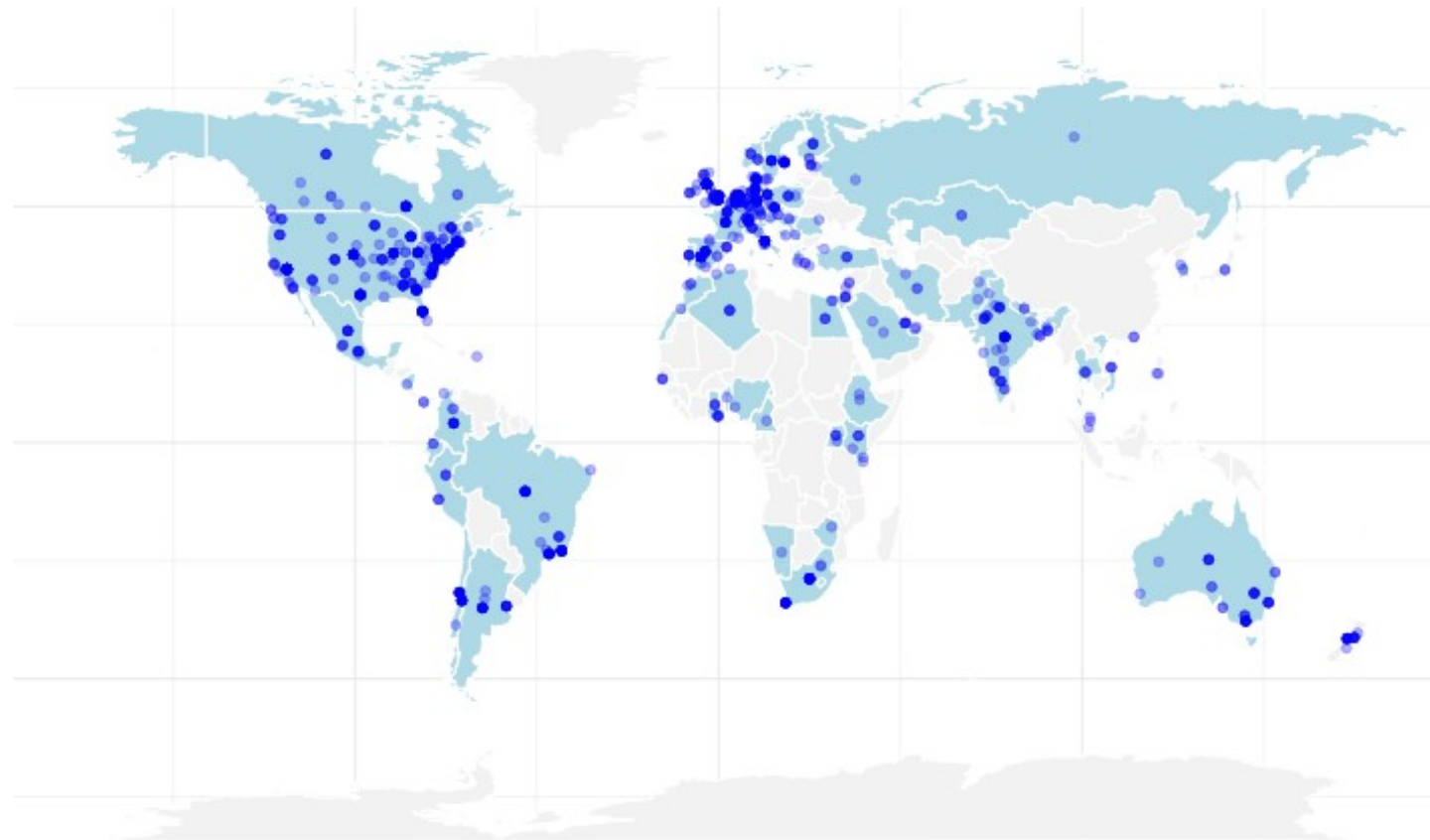
- ❑ Few resources to begin with
- ❑ Those that exist aimed at those with intermediate-advanced R skills, not beginners
- ❑ Know-how concentrated at a few institutions
- ❑ Most self-learning attempts end in frustration



Cytometry in R: A free weekly course for flow cytometrist with no-to-little coding experience



Website



Overall structure



- 30-week course, starting from Intro to R, gradually moving to more advanced content.
- Multiple ways to learn with flexible scheduling
- Taught in Positron instead of Rstudio (ask me why!)
- Version Control/Git introduced from the beginning

YouTube

UMGCCC FCSR
@CytometryInR • 848 subscribers • 21 videos
University of Maryland Greenebaum Comprehensive Cancer Center Flow Cytometry Shared Resource
umgcccfcscr.github.io/CytometryInR
Subscribed

Live Playlists Posts

Latest Popular Oldest

PhD Dissertation Defense - David Rach
Scheduled for 6/17/26, 12:00 PM
Notify me

11 - Data for Statistics
Cytometry Core • 11 - Data for Statistics
AUTHOR David Rach
PUBLISHED May 5, 2026
101 views • Streamed 3 weeks ago

10 - Downsampling and Concatenation
Cytometry Core • 10 - Downsampling and Concatenation
AUTHOR David Rach
PUBLISHED April 27, 2026
129 views • Streamed 1 month ago

09 - It's Raining Functions!
Intro to R • 09 - It's Raining Functions!
AUTHOR David Rach
PUBLISHED April 13, 2026
172 views • Streamed 1 month ago

08 - Manual and Automated Gating
Intro to R • 08 - Manual and Automated Gating
AUTHOR David Rach
PUBLISHED April 7, 2026
226 views • Streamed 1 month ago

07 - Applying Transformations
Intro to R • 07 - Applying Transformations
AUTHOR David Rach
PUBLISHED March 26, 2026
228 views • Streamed 2 months ago

06 - Visualizing with ggplot2
Intro to R • 06 - Visualizing with ggplot2
AUTHOR David Rach
PUBLISHED March 17, 2026
228 views • Streamed 2 months ago

05 - Gating Sets
Intro to R • 05 - Gating Sets
AUTHOR David Rach
PUBLISHED March 1, 2026
253 views • Streamed 2 months ago

- Three live-streams a week (accommodating different timezones and schedules)
- Recordings available immediately after
- Best of the weekly livestreams added to the playlist, cutting down on editing

Website

- ❑ Created with Quarto, hosted via GitHub pages.
- ❑ Code, code outputs, screenshots and explanations
- ❑ Searchable
- ❑ New weekly material added as a new .qmd file

The screenshot displays the 'Cytometry i...' website. The top navigation bar includes links for 'About', 'Schedule', 'Course', 'Announcements', 'Existing Resources', 'Discussions', and 'Homeworks'. A light blue banner at the top of the main content area reads: 'See Course tab for Course Materials, see YouTube tab for livestream recordings'. The left sidebar menu lists the following items: 'Getting Started', 'Intro to R', '01 - Installing R Packages', '02 - File Paths', '03 - Inside a .FCS file', '04 - Intro to Tidyverse', '05 - Gating Sets', '06 - Visualizing With ggplot2', '07 - Applying Transformations', '08 - Ways to Gate' (highlighted in blue), '09 - It's Raining Functions!', 'Cytometry Core', 'Beyond the Sandbox', 'The World is your Oyster', 'Bonus Content', and 'Community'. The main content area is titled 'Lets recreate our GatingSet object:' and contains R code for creating a GatingSet and applying a gating strategy. Below the code, it says 'A GatingSet with 6 samples' and 'And now, let's apply the GatingTable scheme. As we create and click done, the next gate will automatically pop up in the Shiny App.' Below this is a 'Draw your gate' section with a 'Rect' button and a 'Done' button. The 'Gate Type' section has radio buttons for 'Rectangle', 'Polygon' (selected), 'Span', and 'Quadrant', and a checkbox for 'Enable Manual Coord?'. The 'X Minimum' is set to -1000, 'X Maximum' is 50000, and 'Y Minimum' is -1000. To the right of the input fields is a flow cytometry plot showing a population of cells with a rectangular gate drawn around a subset. The plot is titled 'Gate Name: singlets' and 'subset of: root'. The right-hand navigation pane lists the following items: 'Background', 'Walk Through', 'Setup', 'Manual - flowGate', 'Polygon', 'Rectangle', 'Span', 'Quadrant', 'Redraw Gate', 'Saving Gates', 'Rapid Gate Deployment.' (highlighted in blue), 'Visualizing Gates', 'Summary', 'Automated - openCyto', 'Take Away', 'Additional Resources', and 'Take-home Problems'.

GitHub

The screenshot shows the GitHub repository page for 'CytometryInR' by 'UMGCCFCFSR'. The repository is public and has 11 watchers, 603 forks, and 388 stars. The main branch is 'main'. The repository contains a file tree with folders like 'course', 'docs', and 'images', and files like '.gitignore', 'Announcements.qmd', 'CODE_OF_CONDUCT.md', 'CONTRIBUTING.md', 'ExistingResources.qmd', 'Homeworks.qmd', 'LICENSE', 'LICENSE_CC-BY-SA', 'PackageWalkthroughs.qmd', 'README.md', 'Schedule.qmd', '_quarto.yml', and 'index.qmd'. The 'About' section includes a 'Course Website' link, a 'Readme', and a 'Code of conduct'. The 'Contributors' section lists 'DavidRach' and 'UMGCCFCFSR'. The 'Languages' section shows a bar chart with R (85.4%), SCSS (12.4%), JavaScript (1.5%), and GLSL (0.7%).

File/Folder	Description	Last Commit
course	Homework folders week 10 and 11 (thanks Lynn...)	2 days ago
docs	Homework folders week 10 and 11 (thanks Lynn...)	2 days ago
images	Community Package Walkthrough for TRU-OLS	last week
.gitignore	Start of course material transfer	5 months ago
Announcements.qmd	Week 11 slides	last month
CODE_OF_CONDUCT.md	Create CODE_OF_CONDUCT.md	5 months ago
CONTRIBUTING.md	Create CONTRIBUTING.md	5 months ago
ExistingResources.qmd	Existing Resources Links	2 months ago
Homeworks.qmd	Week 11 slides	last month
LICENSE	Start of course material transfer	5 months ago
LICENSE_CC-BY-SA	Start of course material transfer	5 months ago
PackageWalkthroughs.qmd	Updated link to Week #3 recording	4 months ago
README.md	Downsampling - Week 10. Slides and Concatena...	2 months ago
Schedule.qmd	Homework folders week 10 and 11 (thanks Lynn...)	2 days ago
_quarto.yml	Homework folders week 10 and 11 (thanks Lynn...)	2 days ago
index.qmd	Updated in-person schedule, removed course st...	4 months ago

- ❑ Code and datasets available via the GitHub repository
- ❑ Course participants start course by creating GitHub account and forking the repository
- ❑ After Positron is linked to GitHub account, pull locally
- ❑ When new material posted each week, sync and pull



Discussion Forum

- GitHub Discussions page has been repurposed as a community forum for questions, etc.

The screenshot shows the GitHub Discussions interface for the UMGCCCFCSR repository. At the top, there are navigation tabs: Discussions (selected), Actions, Security and quality, Insights, and Settings. Below these are four featured discussion cards: 'Introductions', 'Week # 11 - General Discussion Thread', 'Cytometry in R - Week # 11 - Data for Statistics', and 'Cytometry in R - Course Update and Bonus...'. A search bar with 'is:open' and filters for 'Sort by: Latest activity', 'Label', and 'Filter: Open' are present. The main content area is divided into 'Categories' and 'Discussions'. The 'Categories' sidebar on the left includes 'View all discussions', 'Announcements', 'General', 'Ideas', 'Introductions', 'Polls', 'Q&A', and 'Show and tell'. Below this is a 'Most helpful' section for the last 30 days, featuring a post by 'mpatgaonkar-coder'. The 'Discussions' list on the right shows various topics, including 'Cytometry in R - Week # 07 - Applying Transformations', 'Questions about tidy data', 'Pull error on Week 06 branch', 'Copying index file', 'Week #1 - General Discussion Thread', 'Cytometry in R - Week # 6 - Visualizing with ggplot2', 'GitHub personal access token', 'clock appears when trying to pull from git & commit', 'Bioconductor Cytometry R Packages', and 'No Class This Week'. Each discussion entry includes a lock icon, a title, a user profile, a date, and a comment count.

ns/categories/announcements



Where to next?

Schedule	Introduction to R								Cytometry Core						
															
	Installing R Packages	File Paths	Inside an .FCS file	Intro to Tidyverse	Gating Sets	Visualizing with ggplot2	Applying Transformations	Manual & Automated Gating	It's Raining Functions!	Downsampling & Concatenation	Data for Statistics	Spectral Signatures	Similarities & Hotspots	Unmixing in R	Cleaning Algorithms
	Unsupervised Analysis				Beyond the Sandbox						The World's your Oyster				
															
Clustering Algorithms	Normalization Algorithms	Dimensionality Visualization	Annotating Unsupervised Clusters	The Art of GitHub Diving	XML Files All The Way Down	Utilizing Bioconductor Packages	Building your First R Package	Everyone gets a Quarto Website	Reproducibility & Replicability	Open Source Licenses	Validating Algorithmic Tools	Databases & Repositories	Assembling Web Data	Future Directions	

- ❑ Wrapping up Cytometry Core module
- ❑ Gathering steam to start on the Unsupervised Analysis section

Acknowledgements



ISAC RECOGNIZED
Shared Resource Laboratory

All mothers and their infants, and
their communities

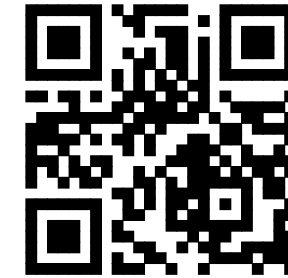
All the open-source contributors,
developers, and maintainers, that
collectively made the tool sets
without which all of this would
not be feasible

Dr. Cristiana Cairo
Dr. Kirsten E. Lyke
Dr. Xiaoxuan Fan

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Cytometry
Discord



Bioconductor
Zulip

 <https://github.com/DavidRach>

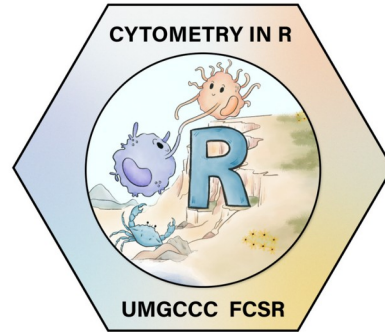
 @davidrach.bsky.social



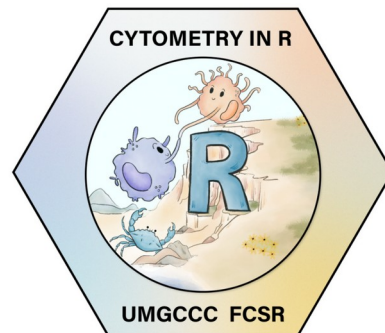
KAMUZU
UNIVERSITY
OF HEALTH SCIENCES

Questions?

Current Works In Progress



Website:
<https://github.com/UMGCCCFCR/CytometryInR>



YouTube @CytometryInR



Luciérnaga



Coereba

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Paper



OPEN ACCESS

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Cord blood innate-like T cell responses in neonates born to healthy women and women living with HIV

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